

Impedance Learning for Human-Guided Robots in Contact With Unknown Environments

Xueyan Xing^{ID}, Etienne Burdet^{ID}, *Member, IEEE*, Weiyong Si, Chenguang Yang^{ID}, *Senior Member, IEEE*, and Yanan Li^{ID}, *Senior Member, IEEE*

Abstract—Previous works have developed impedance control to increase safety and improve performance in contact tasks, where the robot is in physical interaction with either an environment or a human user. This article investigates impedance learning for a robot guided by a human user while interacting with an unknown environment. We develop automatic adaptation of robot impedance parameters to reduce the effort required to guide the robot through the environment, while guaranteeing interaction stability. For nonrepetitive tasks, this novel adaptive controller can attenuate disturbances by learning appropriate robot impedance. Implemented as an iterative learning controller, it can compensate for position dependent disturbances in repeated movements. Experiments demonstrate that the robot controller can, in both repetitive and nonrepetitive tasks: first, identify and compensate for the interaction, second, ensure both contact stability (with reduced tracking error) and maneuverability (with less driving effort of the human user) in contact with real environments, and third, is superior to previous velocity-based impedance adaptation control methods.

Index Terms—Human–robot–environment interaction control, impedance control, learning control.

NOMENCLATURE

x	Position of the robot end-effector.
x_e	Rest position of the environment.
d	Environmental disturbance.

Manuscript received 23 December 2022; accepted 28 April 2023. Date of publication 23 June 2023; date of current version 4 October 2023. This work was supported in part by the U.K. EPSRC under Grant EP/T006951/1, in part by the (Chinese) State Key Laboratory of Robotics and Systems (HIT) under Grant SKLRS-2022-KF-16, and in part by the European Commission under Grant ICT-871803 CONBOTS. This paper was recommended for publication by Associate Editor D. Lee and Editor P. R. Giordano upon evaluation of the reviewers' comments. (*Corresponding author: Yanan Li.*)

This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by Sciences and Technology Cross-Schools Research Ethics Committee of Sussex University, and the UWE Faculty Research Ethics Committee under Application No. ER/YL557/1, and FET-2122-59, respectively.

Xueyan Xing and Yanan Li are with the Department of Engineering and Design, University of Sussex, BN1 9RH Brighton, U.K. (e-mail: xx92@sussex.ac.uk; yl557@sussex.ac.uk).

Etienne Burdet is with the Department of Bioengineering, Imperial College of Science, Technology and Medicine, SW7 2AZ London, U.K. (e-mail: eburdet@ic.ac.uk).

Weiyong Si and Chenguang Yang are with the Bristol Robotics Laboratory, University of the West of England, BS16 1QY Bristol, U.K. (e-mail: weiyong.si@uwe.ac.uk; cyang@ieee.org).

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TRO.2023.3281483>.

Digital Object Identifier 10.1109/TRO.2023.3281483

K, K_v, K_e

Coefficients of the disturbance and control matrix.

$K_s, K_{vs}, k_{ps}, x_{es}$
 A_f, A_d, A_c

Spatial form of K, K_v, k_p, x_e .

Parameters of the proposed adaptive impedance learning method.

v

Velocity of the robot end-effector.

V

Lyapunov function of the system (for stability analysis).

E

Total tracking error.

M, B, g

Inertia, Coriolis coefficient, and gravity of the robot.

x_h, x_r

Reference positions of the human and robot.

u

Robot control input.

$\hat{K}, \hat{K}_v, \hat{k}_p$

Estimations of K, K_v, k_p , where $k_p = -Kx_e$.

$\hat{K}_s, \hat{K}_{vs}, \hat{k}_{ps}$
 A, A_v, A_p, Λ

Estimations of K_s, K_{vs}, k_{ps} .

Parameters of the proposed iterative impedance learning method.

ξ

Integration variable of space (for stability analysis).

a_j, b_j, c_j

Parameters of the velocity-based adaptation in (24).

P_E

Capability of the method in terms of contact stability.

K_h, K_{hv}

Stiffness and viscosity matrices of the human arm.

f_h

Interaction force between the human and robot.

e_h, e

Tracking errors of the human and robot.

T_i

i th iteration period for repetitive tasks.

$\overline{T}$

Maximum iteration period for repetitive tasks.

η, μ

Parameters of the improved Λ .

σ

Integration variable of time (for stability analysis).

J

Index of driving effort of the human user.

P_F

Capability of the method in terms of maneuverability.

I. INTRODUCTION

VARIOUS robotic applications, such as (tele)operating a microrobot, demand a robot to be guided by a human while interacting with the environment. Most existing robotic

interaction control schemes focus on achieving an effective interaction either with the human operator, or with an environment. However, during the execution of many human–robot interaction tasks, the robot may concurrently interact with an environment, for example, during human-guided tooling. This interaction may destabilize the movement, affect task performance, and fatigue the human operator. Our vision to avoid these shortcomings is for a robot to understand the human intention, while learning to compensate for identified reproducible interaction dynamics during movement.

When a human user guides a robot to complete a tooling task, such as cutting, polishing, or drilling, they need to exert a force on the robot to move it along the planned trajectory and achieve satisfying tooling performance. For instance, as a human user carries out a carving task, they need to apply a certain force to manipulate the blade along a desired trajectory. If the engraved plate is of hard texture, there will be a large resistive disturbance that will degrade the carving performance. If the human user wants to maintain the carving quality, more effort is needed to overcome the unknown disturbance. In this case, a robot that complies with human movement while concurrently attenuating environmental disturbance is required.

To analyze the interaction of the human operator with the robot and its environment, we define *maneuverability* as the degree of difficulty to guide the robot for completing a task in the presence of environmental perturbations. The more easily the human operator can guide the robot, the higher the maneuverability. We also define *contact stability* as the capability for the robot to ensure good task performance despite environmental perturbations. The robot should both i) allow the human operator to easily move and flexibly operate the robot, thus yielding high maneuverability, and ii) complete the task with a small error in the presence of perturbation from the environment, such as vibration due to a nonhomogeneous material and undesirable friction, i.e., the contact stability has to be guaranteed. Usually, the contact stability is guaranteed at the expense of the maneuverability, or vice versa. When the human user tries to improve their performance, i.e., to reduce the tracking error, they need more effort to compensate for the disturbance, which will result in decreasing maneuverability. Conversely, with less human control effort, the task performance will be deteriorated due to the negative effects of the disturbance. In other words, maneuverability and contact stability are hard to ensure concurrently. Our goal is to develop an interaction controller that makes robots able to interact with their environment, while offering both high maneuverability and contact stability to the human operator.

A. Previous Works in Human–Robot Interaction

How to design a robot controller for *intuitive and efficient human–robot interaction* is a prevalent research topic in the robotics field. Robots that interact with humans to perform their functions include rehabilitation robots, robotic exoskeletons, collaborative robots (cobots) for micro manipulation and industrial assembly, and semiautonomous vehicles. Impedance control is often used on these contact robots to yield a compliant interaction of a robot with its human operator.

Facing uncertain human behaviors, a number of works estimate the human impedance or intention to dynamically adjust the robot's impedance parameters for efficient and stable interaction. For instance, it has been proposed to regulate the robot impedance using muscle activity measured with surface electromyography, although the underlying control is limited by the large signal variability and neural constraints [1], [2]. To address these limitations, several approaches used the force and position information of the human limb, which can be directly measured by sensors, to infer their impedance and adjust the robot's impedance correspondingly [3], [4]. Other works estimate intention or characteristics of the human partner, e.g., trajectory in [5] or human impedance in [6] and [7] for adjustment of the robot's impedance. Alternatively, a velocity-related heuristic rule scheme has been considered in [4] and [8] to adjust robot impedance, rather than relying on the human partner's behavior. In addition, several works considered that human movements obey a certain probability distribution that can be identified to infer their intention [9]. Based on time-series data, the motion intention was estimated using the sparse Bayesian learning method in [10], and an adaptive impedance control algorithm was developed for effective human–robot collaboration. Instead of adjusting the robot's impedance, the robot's reference trajectory can be updated to achieve efficient human–robot interaction [11]. A review of human–robot interaction control can be found in [12]. However, previous works do not explicitly consider concurrent interactions between the robot, human operator, and environment.

B. Existing Works on Robot–Environment Interaction

Contact tooling tasks by definition involve *a desired interaction between the robot and the environment*. Control theory has been used to consider robot–environment interaction stability, and to develop impedance adaptation. In [13], [14], and [15], the interaction force between the robot and the human was considered as an environmental force, where neural and iterative learning controls were used to mitigate its negative effect. Different from [13], [14], and [15], the adaptive control strategy in [16] could generate the desired interaction force between robot and environment by adjusting the feedforward force, impedance, and trajectory of the robot. Reinforcement learning approaches have also been proposed to determine the optimal impedance parameters of the robot controller to achieve desired robot–environment interaction performance [17], [18], [19], [20]. A reward function including contact information, e.g., the interaction force, was used to determine the optimal impedance parameters of the robot controller to achieve desired robot–environment interaction performance [21]. Fuzzy logic controllers have also been developed for impedance adaptation of the robot. Based on a fuzzy neural network learning algorithm, an adaptive controller was designed in [22] for adjusting the impedance parameters of the robot so as to produce smooth control even during transitions from free movement to contact. However, the controllers designed in [13], [14], [15], [16], [17], [18], [19], [20], [21], and [22] do not consider a human operator.

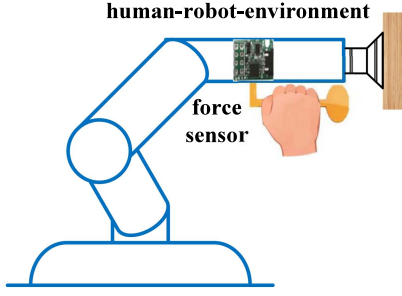


Fig. 1. Human-robot-environment interaction: A robot is guided by a human operator while in physical contact with an environment.

C. Contributions of This Article on Human-Robot-Environment Interaction

Different from the aforementioned works that considered either the interaction between robot and environment, or the interaction between robot and human, we will consider here both the human operator and the environment. We will design a robot controller that automatically infers the intention of the human and adapts to an uncertain environment by learning the resulting impedance to improve the task performance. It is not a simple combination of two controllers for robot-environment interaction and human-robot interaction, as the two control objectives can be contradictory, leading to a conflicting trade-off between maneuverability and contact stability as discussed above. Our idea is to maintain the robot's maneuverability by inferring the human intent while only introducing additional impedance to deal with the environmental disturbance and, thus, ensure contact stability. As a result, both the maneuverability and the contact stability can be ensured during the human-robot-environment interaction, and the conflict between the human effort and the task performance is addressed.

The designed impedance learning strategy can be applied to both nonrepetitive tasks, such as engraving or drawing, in the form of adaptive learning, and repetitive tasks, such as grinding or polishing, in the form of iterative learning. In the collaborative tasks considered in this article, the robot is guided by the human user that assists them to complete a task in the presence of uncertain environmental disturbances. As illustrated by Fig. 1, the human user holds the robot and their interaction force can be measured by a force sensor. Simultaneously, the robot interacts with the environment for a certain tooling task. As the interaction force between the robot and the environment is typically subject to significant noise, it is not measured by a force sensor. Instead, due to repetitive interaction and periodicity of the environmental parameters, the disturbance from the environment can be learned by the robot.

The rest of this article is organized as follows. Section II formulates dynamics of and issues with the human-guided robot in contact with the environment. An adaptive variable impedance control is developed in Section III for nonrepetitive tasks. Section IV presents an iterative control method with variable impedance learning for repetitive tasks considering position dependent disturbances. An experiment to test the algorithm's ability to identify a known virtual environment and

yield stable interaction behavior is described in Section V. Section VI presents an experiment to test the effectiveness of the proposed impedance learning scheme in interaction with real environments. Finally, Section VII concludes this article and analyzes the advantage of the proposed control method by comparing with the existing velocity-based adaptation method. The variables used in this article are listed in the Nomenclature.

II. PROBLEM FORMULATION

A. Dynamics of Human-Robot-Environment Interaction

We aim to design a controller for a human-guided robot that can stably interact with an environment. The dynamics of the n -degree of freedom (DoF) human-guided robot can be modeled as

$$M \ddot{x} + B \dot{x} + g = f_h + d + u \quad (1)$$

where $x = [x_1, \dots, x_n]^T \in \mathbb{R}^n$ is the position vector of the end-effector, $M \in \mathbb{R}^{n \times n}$ is the inertia matrix of the robot, $B \in \mathbb{R}^{n \times n}$ is the positive definite matrix related to the Coriolis force, $g \in \mathbb{R}^n$ is the gravity force on the robot, $f_h \in \mathbb{R}^n$ is the vector of the measured interaction force signals, $d \in \mathbb{R}^n$ is the disturbance due to the interaction with an unknown environment, and $u \in \mathbb{R}^n$ is the robot command vector. In the model in (1), parameters of the robot M, B, g are time-varying and state-related, but their arguments are omitted in this article to facilitate the readability.

In this article, the human force f_h is expected to be minimized, even with an environmental disturbance d . While both f_h and d are external forces on the robot controller, f_h is considered “desirable” as it indicates the human's planned trajectory, and d is “undesirable” as it may cause instability.

The interaction force can be modeled as [23]

$$f_h = M \ddot{x}_h + B \dot{x}_h - K_h e_h - K_{hv} \dot{e}_h, e_h = x - x_h \quad (2)$$

where $x_h \in \mathbb{R}^n$ is the desired position vector of the human that is unknown to the robot, $K_h \in \mathbb{R}^{n \times n}$ and $K_{hv} \in \mathbb{R}^{n \times n}$ are stiffness and viscosity matrices of the human arm, and $e_h \in \mathbb{R}^n$ is the tracking error between x_h and x . The first two terms of (2) correspond to the feedforward force to compensate for the system dynamics and the last two terms to muscle viscoelasticity. Notice that (2) is not used in the following robot controller design, but only for analysis purpose. If the robot is without disturbance and only with gravity compensation, i.e., $d = 0$ and $u = g$, then combining (1,2) yields

$$M \ddot{e}_h + (B + K_{hv}) \dot{e}_h + K_h e_h = 0 \quad (3)$$

which shows that the robot can stably follow x_h corresponding to the human's motion intent and the instability in practice is largely caused by the environmental disturbance.

Since the human force can guide the robot to the desired position without disturbance and control, under such an ideal case, the robot passively follows the desired human trajectory as expected and $e_h \approx 0$. Then, using (2), we have

$$M \ddot{x}_h + B \dot{x}_h = f_h. \quad (4)$$

If we define the reference position vector of the robot as $\mathbf{x}_r \in \mathbb{R}^n$, replacing $\mathbf{x}_h$ in (4) with $\mathbf{x}_r$ yields

$$\mathbf{M} \ddot{\mathbf{x}}_r + \mathbf{B} \dot{\mathbf{x}}_r = \mathbf{f}_h. \quad (5)$$

With (5), the reference trajectory of the robot $\mathbf{x}_r$ is available for the impedance control designed in the following sections.

B. Stability Analysis for Human–Robot–Environment Interaction

This section analyzes the system stability to develop an impedance learning controller for an ideal case with measurable disturbance. Let us consider a general viscoelastic model of the environment:

$$\mathbf{d} = \mathbf{K}(\mathbf{x} - \mathbf{x}_e) + \mathbf{K}_v \dot{\mathbf{x}} \quad (6)$$

where $\mathbf{K} \in \mathbb{R}^{n \times n}$ and $\mathbf{K}_v \in \mathbb{R}^{n \times n}$ are the coefficients of the disturbance and $\mathbf{x}_e \in \mathbb{R}^n$ is the rest position of the environment.

When there is no external disturbance and the robotic control is designed only for gravity compensation, i.e., $\mathbf{u} = \mathbf{g}$ and $\mathbf{d} = \mathbf{0}$, according to (1)

$$\mathbf{M} \ddot{\mathbf{x}} + \mathbf{B} \dot{\mathbf{x}} = \mathbf{f}_h \quad (7)$$

in which $\mathbf{f}_h$ is driving the robot toward $\mathbf{x}_h$ as shown in (4) and the interaction system is stable if (7) holds.

If only the gravity compensation is considered in the robotic controller, considering the system in (1) with $\mathbf{u} = \mathbf{g}$ and the disturbance $\mathbf{d}$ in (6), the model of the human-guided robot can be rewritten as

$$\mathbf{M} \ddot{\mathbf{x}} + \bar{\mathbf{B}} \dot{\mathbf{x}} + \bar{\mathbf{C}} \mathbf{x} + \bar{\mathbf{a}} = \mathbf{f}_h \quad (8)$$

where

$$\bar{\mathbf{B}} = \mathbf{B} - \mathbf{K}_v, \quad \bar{\mathbf{C}} = -\mathbf{K}, \quad \bar{\mathbf{a}} = \mathbf{K} \mathbf{x}_e. \quad (9)$$

It is clear that when the disturbance is resistive to the robot movement, i.e., the matrices $\mathbf{K}$ and $\mathbf{K}_v$ are positive definite, the human-guided robot in (8) could become unstable in the presence of large disturbances because of the undesired terms $\bar{\mathbf{B}}$, $\bar{\mathbf{C}} \mathbf{x}$, and $\bar{\mathbf{a}}$ in (9), such that more human effort is needed for a stable contact. Therefore, to achieve a stable contact with the environment while maintaining the robot's maneuverability, a robotic controller is required to attenuate the effect of the disturbance.

According to the model in (1), if the environmental disturbance $\mathbf{d}$ is measurable, a robot controller with gravity compensation can be designed as

$$\mathbf{u} = -\mathbf{d} + \mathbf{g} \quad (10)$$

such that (7) holds even with the disturbance. This would enable the human to use little effort to ensure the contact stability and guide the robot. In practical operation, the environmental disturbance is hard to be accurately estimated. In the following two sections, an impedance controller is designed based on the human-guided robot model in (1), which does not require knowledge of $\mathbf{d}$ and $\mathbf{x}_h$ to guarantee the contact stability and the robot maneuverability.

III. IMPEDANCE CONTROL DESIGN FOR NONREPETITIVE TASKS

For the nonrepetitive task considered in this section, the coefficients $\mathbf{K}$ and $\mathbf{K}_v$ of the disturbance in (6) and the rest position of the environment $\mathbf{x}_e$ are constant. Then, an adaptive impedance control is designed as

$$\mathbf{u} = -\hat{\mathbf{K}} \mathbf{x} - \hat{\mathbf{K}}_v \dot{\mathbf{x}} - \hat{\mathbf{k}}_p - \mathbf{K}_e \dot{\mathbf{e}} + \mathbf{g} \quad (11)$$

with impedance learning laws

$$\dot{\hat{\mathbf{K}}} = \dot{\mathbf{e}} \mathbf{x}^T \mathbf{A}_f, \quad \dot{\hat{\mathbf{K}}}_v = \dot{\mathbf{e}} \dot{\mathbf{x}}^T \mathbf{A}_d, \quad \dot{\hat{\mathbf{k}}}_p = \mathbf{A}_c \dot{\mathbf{e}} \quad (12)$$

where $\mathbf{e} \in \mathbb{R}^n$ is the tracking error of the robot

$$\mathbf{e} = \mathbf{x} - \mathbf{x}_r \quad (13)$$

and $\mathbf{A}_f$, $\mathbf{A}_d$, $\mathbf{A}_c$, $\mathbf{K}_e \in \mathbb{R}^{n \times n}$ are positive definite diagonal matrices.

Equation (12) can be used to compute $\hat{\mathbf{K}}$, $\hat{\mathbf{K}}_v$, and $\hat{\mathbf{k}}_p$ for the control of (11). For example, $\hat{\mathbf{K}}$ can be computed from $\hat{\mathbf{K}}(t + \Delta t) = \dot{\mathbf{e}} \mathbf{x}^T \mathbf{A}_f \Delta t + \hat{\mathbf{K}}(t)$ using Euler integration with a predefined initial value $\hat{\mathbf{K}}(0)$, where Δt denotes the sampling time. In (12), we know that $\hat{\mathbf{K}}$, $\hat{\mathbf{K}}_v$, and $\hat{\mathbf{k}}_p$ update depending on the variation speed of the tracking error $\dot{\mathbf{e}}$, and the position $\mathbf{x}$ and velocity $\dot{\mathbf{x}}$ of the system. Since the disturbance $\mathbf{d}$ in (6) can be rewritten as $\mathbf{d} = \mathbf{K} \mathbf{x} - \mathbf{K} \mathbf{x}_e + \mathbf{K}_v \dot{\mathbf{x}}$, it is related to the position $\mathbf{x}$, the velocity $\dot{\mathbf{x}}$ of the system, as well as the rest position of the environment $\mathbf{x}_e$. As a result, the controller will update its three parameters according to the current tracking performance, which is indicated by $\dot{\mathbf{e}}$, and the current degree of disturbance, which is indicated by $\mathbf{x}$ and $\dot{\mathbf{x}}$. Besides, according to the backward difference method, $\dot{\mathbf{x}}_r$ can be represented as

$$\dot{\mathbf{x}}_r = \frac{\mathbf{x}_r(t) - \mathbf{x}_r(t - \Delta t)}{\Delta t},$$

where Δt is the sampling time, and thus,

$$\ddot{\mathbf{x}}_r = \frac{\mathbf{x}_r(t) - 2\mathbf{x}_r(t - \Delta t) + \mathbf{x}_r(t - 2\Delta t)}{(\Delta t)^2}$$

according to [24]. By substituting the expressions of $\dot{\mathbf{x}}_r$ and $\ddot{\mathbf{x}}_r$ into (5), the reference trajectory of the robot $\mathbf{x}_r(t)$ used in (13) can be updated using $\mathbf{x}_r(t) = [(\Delta t)^2 \mathbf{f}_h - \mathbf{M} \mathbf{x}_r(t - 2\Delta t) + (\mathbf{B} \Delta t + 2\mathbf{M}) \mathbf{x}_r(t - \Delta t)] \times (\mathbf{M} + \mathbf{B} \Delta t)^{-1}$.

Substituting (11) into (1) yields

$$\mathbf{M} \ddot{\mathbf{x}} + (\mathbf{B} + \hat{\mathbf{K}}_v) \dot{\mathbf{x}} + \hat{\mathbf{K}} \mathbf{x} + \mathbf{K}_e \dot{\mathbf{e}} + \hat{\mathbf{k}}_p = \mathbf{f}_h + \mathbf{d} \quad (14)$$

which can be rewritten as

$$\mathbf{M} \ddot{\mathbf{x}} + \tilde{\mathbf{B}} \dot{\mathbf{x}} + \tilde{\mathbf{C}} \mathbf{x} + \mathbf{K}_e \dot{\mathbf{e}} + \tilde{\mathbf{a}} = \mathbf{f}_h + \mathbf{d} \quad (15)$$

where

$$\tilde{\mathbf{B}} \equiv \mathbf{B} + \hat{\mathbf{K}}_v, \quad \tilde{\mathbf{C}} \equiv \hat{\mathbf{K}}, \quad \tilde{\mathbf{a}} \equiv \hat{\mathbf{k}}_p. \quad (16)$$

From (16), it is clear that the impedance matrices $\tilde{\mathbf{B}}$, $\tilde{\mathbf{C}}$, $\tilde{\mathbf{a}}$ are regulated according to the variation of the disturbance $\mathbf{d}$ with learning laws in (12). When $\hat{\mathbf{K}} \mathbf{x} + \hat{\mathbf{K}}_v \dot{\mathbf{x}} + \hat{\mathbf{k}}_p \simeq \mathbf{K} \mathbf{x} + \mathbf{K}_v \dot{\mathbf{x}} - \mathbf{K} \mathbf{x}_e$, (7) holds with an additional assistive force $\mathbf{K}_e \dot{\mathbf{e}}$ provided by the robot such that the human can guide the robot with $\mathbf{f}_h$ free from the disturbance.

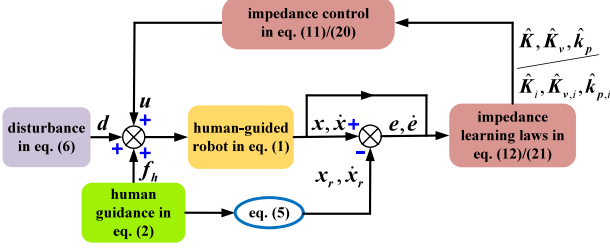


Fig. 2. Control block diagram of the human-guided robot for nonrepetitive and repetitive tasks.

Using (5), (6), (11), eq. (1) can be rewritten as

$$M \ddot{e} + (B + K_e) \dot{e} = \tilde{K} x + \tilde{K}_v \dot{x} + \tilde{k}_p \quad (17)$$

where $\tilde{K}, \tilde{K}_v \in \mathbb{R}^{n \times n}$, and $\tilde{k}_p \in \mathbb{R}^n$ are defined as

$$\tilde{K} = K - \hat{K}, \tilde{K}_v = K_v - \hat{K}_v, \tilde{k}_p = k_p - \hat{k}_p \quad (18)$$

while $k_p = -K x_e$.

Using these notations, we can formulate Theorem 1: For an n -DoF human-guided robot used in a nonrepetitive task, the adaptive impedance control strategy in (11) and (12) can eliminate the adverse effect of the disturbance d and regulate the robot behavior so that the stability of the closed-loop system in (17), (18) is guaranteed in the presence of d . The proof of Theorem 1 is provided in Appendix A. ■

Theorem 1 indicates that the disturbance can be learned using our adaptive impedance control scheme, thus the controller in (11) enables (7) to hold for the closed-loop interaction system. As impedance learning in (12) regulates the robot impedance to compensate for the effect of the disturbance during the interaction, the human user does not need to overcome the contact instability caused by the disturbance, and the effort to steer the robot is reduced. The abovementioned control process can be found in Fig. 2.

In (11), u is derived by analyzing the stability of the disturbed human–robot–environment interaction system based on Lyapunov's stability theory [25]. We first need to select a Lyapunov function of the system [V in (32) in Appendix A] that includes the tracking error and the estimation error of the disturbance parameters that we want to minimize, then u is designed to make the controlled system stable through $\dot{V} \leq 0$.

For the designed control u , the initial values of the estimated control parameters in (11), $\hat{K}(0), \hat{K}_v(0), \hat{k}_p(0)$, affect the transient performance but not the steady-state performance, and they are typically initialized as zero. However, the learning parameters A_f, A_d, A_e in (12) should be manually tuned to be large enough to provide adequate convergence speed of the controlled system.

IV. ITERATIVE IMPEDANCE CONTROL FOR HUMAN-GUIDED ROBOT IN REPETITIVE TASKS

When the robot performs a repetitive task, the environment has spatial periodicity, thus, the disturbance can be considered as a function of the position as discussed in [26]. To further elaborate, instead of constant matrices, coefficients of the disturbance are spatially periodic, i.e., their values are the same for the same position in different iterations. We modify here the

adaptive impedance control proposed in Section III for repetitive tasks, making it suitable for space-related disturbances with varying coefficients. The subscript s is used for space-related variables and time-related variables are without s . The subscript i of a variable denotes its iteration number.

In the i th iteration, the spacial form of the environmental disturbance d_i in (6) is written as

$$d_i = K_s(x_i - x_{es}) + K_{vs} \dot{x}_i \quad (19)$$

where $K_s, K_{vs} \in \mathbb{R}^{n \times n}$, and $x_{es} \in \mathbb{R}^n$ are the spatial form of K, K_v , and x_e , respectively, which satisfy $K_{s,i} = K_{s,i-1}$, $K_{vs,i} = K_{vs,i-1}$, $x_{es,i} = x_{es,i-1}$.

Based on the disturbance in (19), the robot control u is redesigned as

$$u_i = -\hat{K}_i x_i - \hat{K}_{v,i} \dot{x}_i - \hat{k}_{p,i} - K_e \dot{e}_i + g \quad (20)$$

where the control parameter matrix $K_e \in \mathbb{R}^{n \times n}$ keeps the same for each iteration, and impedance parameter matrices $\hat{K}_i \in \mathbb{R}^{n \times n}$, $\hat{K}_{v,i} \in \mathbb{R}^{n \times n}$, and $\hat{k}_{p,i} \in \mathbb{R}^n$ are updated in time as

$$\begin{aligned} \hat{K}_i &= \begin{cases} \hat{K}_{i-1} + \dot{e}_i x_i^T A \Lambda & \text{for } t \in [0, T_i] \\ \hat{K}_{i-1} & \text{for } t \in (T_i, \bar{T}] \end{cases} \\ \hat{K}_{v,i} &= \begin{cases} \hat{K}_{v,i-1} + \dot{e}_i \dot{x}_i^T A_v \Lambda & \text{for } t \in [0, T_i] \\ \hat{K}_{v,i-1} & \text{for } t \in (T_i, \bar{T}] \end{cases} \\ \hat{k}_{p,i} &= \begin{cases} \hat{k}_{p,i-1} + A_p \Lambda \dot{e}_i & \text{for } t \in [0, T_i] \\ \hat{k}_{p,i-1} & \text{for } t \in (T_i, \bar{T}] \end{cases} \end{aligned} \quad (21)$$

where A, A_v , and A_p are $\mathbb{R}^{n \times n}$ diagonal matrices with positive constant elements, $T_i \in [0, \bar{T}]$ is the actual learning time of the i th iteration, $\bar{T}$ is the maximum iteration period, $\Lambda = \text{diag}\{\lambda^1 \dots \lambda^n\} \in \mathbb{R}^{n \times n}$ in which $\lambda^j = |v^j|^{-1}$ and $v^j \in \mathbb{R}$ is the current robot velocity along the j th dimension, $j = 1, \dots, n$.

In traditional iterative learning control, a fixed iteration period is needed to ensure system stability. However, humans never repeat a movement with exactly the same time duration [27]. In our proposed method, the learning laws in (21) are designed as piecewise functions. Based on this, we use the method in [15] and introduce Λ to deal with the varying iteration period. When the current velocity increases, the learning speed will correspondingly decrease due to the decrease of Λ , so that the possible spatial inconsistency caused by the varying velocity can be corrected with our proposed update laws, and the stability of the human–robot–environment interaction system can be guaranteed. With the maximum iteration period $\bar{T}$, the controller updates its coefficients for $t \in [0, T_i]$ and stops updating for the rest of the duration since there is no valuable information to be learned. $\bar{T}$ could be theoretically set to be large enough to cover all possibilities so that the actual learning time T_i can eventually achieve it in the sense of $\lim_{i \rightarrow \infty} T_i = \bar{T}$ [15], [28]. However, if $\bar{T}$ is too large, the control period will be correspondingly extended, which will reduce the control efficiency. Therefore, $\bar{T}$ should be determined according to the task execution time. In practice, $\bar{T}$ can be set slightly larger than the prescribed movement time (e.g., as 4'30" for a 4' planned duration) to guarantee that the task can be completed.

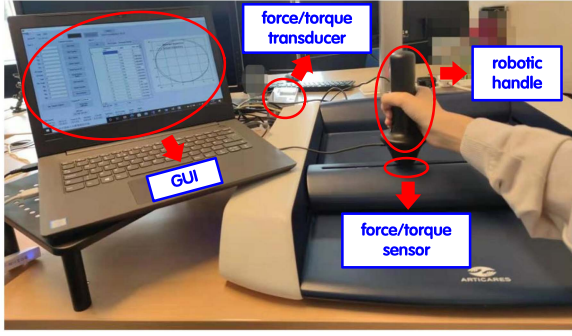


Fig. 3. Experimental setup with H-MAN robot to test the algorithm on a virtual environment interaction.

Substituting (5), (6), (20) into (1) yields

$$M \ddot{e}_i + (B + K_e) \dot{e}_i = \tilde{K}_i x_i + \tilde{K}_{v,i} \dot{x}_i + \tilde{k}_{p,i} \quad (22)$$

where $\tilde{K}_i, \tilde{K}_{v,i} \in \mathbb{R}^{n \times n}$, and $\tilde{k}_{p,i} \in \mathbb{R}^n$ are defined as

$$\tilde{K}_i = K_i - \hat{K}_i, \tilde{K}_{v,i} = K_{v,i} - \hat{K}_{v,i}, \tilde{k}_{p,i} = k_{p,i} - \hat{k}_{p,i} \quad (23)$$

where $k_{p,i} = -K_i x_{e,i}$.

With above-mentioned equations, we can formulate Theorem 2: For the n -DoF human-guided robot described by (1) in repetitive tasks, the iterative impedance control strategy in (20) with impedance learning laws in (21) can guarantee the stability of the closed-loop system in (22) and (23) in the presence of environmental disturbance, i.e., (7) is satisfied with the proposed control. The proof of this theorem is presented in Appendix B, where the robot control is first designed to be space-related to conform with (19) and then transformed to time domain for implementation.

The block diagram of the resulting human-guided robot control for repetitive tasks is shown in Fig. 2. In order to avoid $\hat{K}$, $\hat{K}_v$, and $\hat{k}_p$ becoming too large when the robot velocity is close to zero, λ^j defined in (21) can be revised with $\lambda^j = \begin{cases} |v^j|^{-1}, & \text{if } |v^j| \geq \mu \\ (|v^j| + \eta)^{-1}, & \text{if } |v^j| < \mu \end{cases}$, where a constant η is added to λ^j when $|v^j|$ is less than a positive constant μ .

V. CONTROL VALIDATION WITH H-MAN ROBOT

To test the ability of the proposed impedance controller to identify the environmental disturbances and stably compensate for it, an experiment was implemented on a planar H-MAN robot with a virtual, software designed disturbance. The experiment was approved by the Sciences and Technology Cross-Schools Research Ethics Committee of Sussex University, U.K. (reference number ER/YL557/1). Six human subjects without known sensorimotor impairment were recruited, who all gave their informed consent prior to participating. International units are used in the following, unless specified.

The experimental setup with the planar H-MAN robot is shown in Fig. 3. A graphical user interface (GUI) is designed to display the target trajectory and the cursor indicating the handle's position. In the experiment, the human operator holds the handle

and guides the robot to track the target trajectory with the cursor on the GUI. The H-MAN robot is equipped with an ATI Mini 40 force/torque sensor at the handle. Since the virtual disturbance is introduced in software as a part of the control input of the robot (instead of stemming from interaction with a physical environment), the force measured by the load cell is only from the human.

To evaluate our proposed adaptive impedance control, the following three control conditions were compared.

- 1) *Control off*: The robot control was 0.
- 2) *Velocity-based*: The velocity-based impedance adaptation proposed in [4]

$$u_j = -\min \left\{ a_j e^{-b_j |\dot{x}_j|}, c_j \right\} \dot{x}_j, \quad a_j, b_j, c_j > 0 \quad (24)$$

where $j = 1, 2$ stands for x or y axis.

- 3) *Impedance learning*: The proposed control method with impedance learning was used.

Each subject was instructed on how to use the H-MAN robot, and then practiced the task. Recording began when the subject felt confident in performing the task. Each subject carried out five trials in each of the three control conditions, whose order was unknown to them.

Sections V-A–C present the results of a representative subject, and the overall performance of the six subjects is described in Section V-D.

A. Nonrepetitive Task

In the nonrepetitive task, the target displayed on the GUI moved in a circular trajectory $[10 \cos(\frac{\pi}{4.5}t) \ 10 \sin(\frac{\pi}{4.5}t)]^T$ cm. One trial consisted of tracking this trajectory once. The disturbance was set as $\mathbf{d} = [d_x \ d_y]^T = [20x_1 + 4\dot{x}_1 \ 20x_2 + 4\dot{x}_2]^T$ N.

The velocity-based control in (24) was implemented with $a_1 = a_2 = 500$, $b_1 = b_2 = 3$, and $c_1 = c_2 = 50$. Impedance learning was implemented using the controller in (11), (12) with parameters $K_e = 0.1 I_2$, $A_f = 0.5 I_2$, $A_d = I_2$, $A_c = 0.05 I_2$, where $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

The maneuverability defined in this article was evaluated using the l_2 -norm of the instantaneous tracking error $\|e\|$, while the contact stability was measured using the human effort, J , to drive the robot

$$J = \int_0^{T_0} |\mathbf{f}_h^T(t) \mathbf{v}(t)| dt \quad (25)$$

where T_0 is the task duration and $\mathbf{v}$ is the velocity of the interaction system. To evaluate the disturbance identification, the learned two-dimension disturbance $\hat{\mathbf{d}}$ for nonrepetitive tasks

$$\hat{\mathbf{d}} = [\hat{d}_x \ \hat{d}_y]^T = \hat{K} \mathbf{x} + \hat{K}_v \dot{\mathbf{x}} + \hat{k}_p \quad (26)$$

can be analyzed, where $\hat{K}$, $\hat{K}_v$, $\hat{k}_p$ are coefficients of the disturbance learned according to (12).

Fig. 4(a) shows the trajectories of a representative subject in one trial. In Fig. 4(a), the dashed black line, corresponding to the subject guiding the robot without control and without

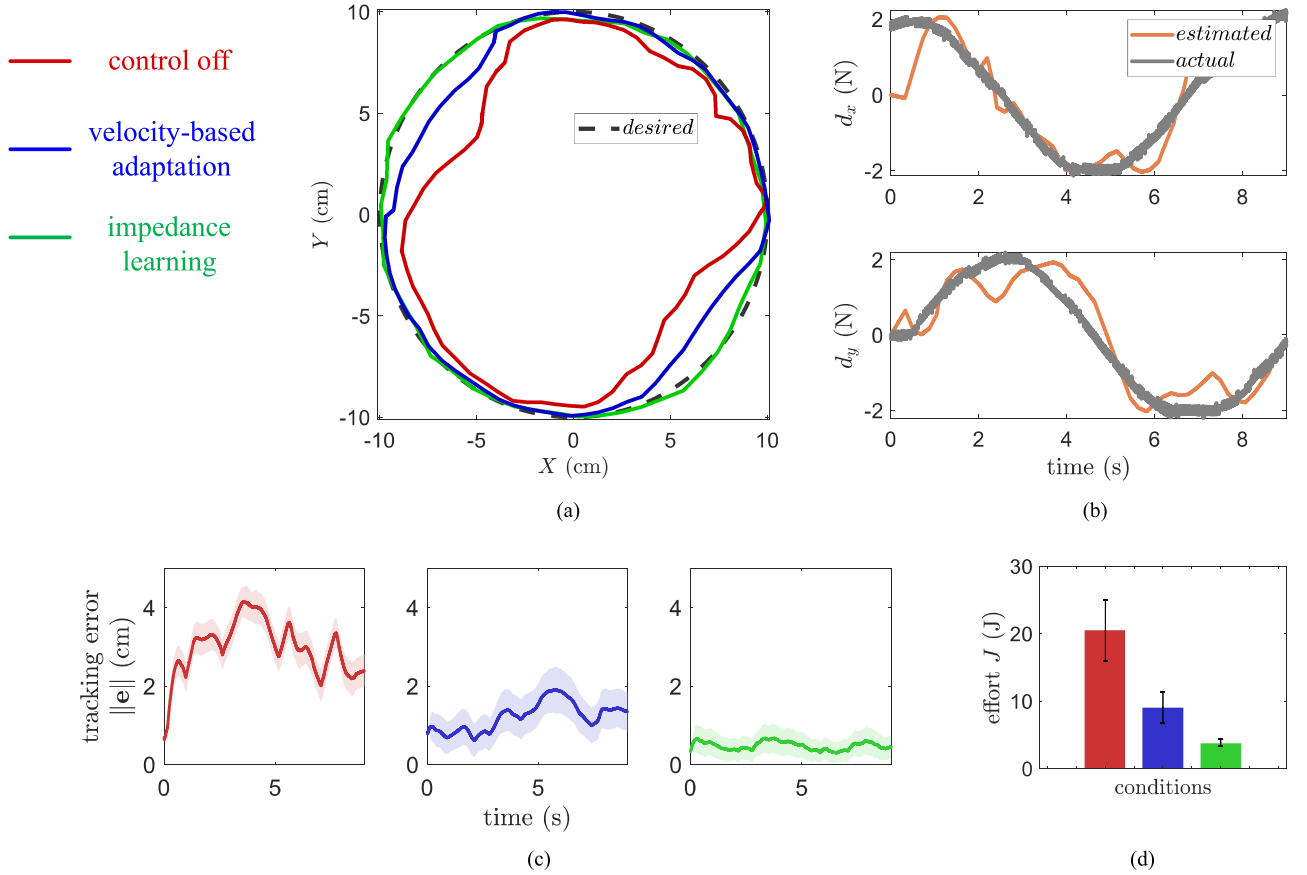


Fig. 4. Results of a representative subject for the nonrepetitive task with the H-MAN robot, in the conditions of control off (in red), velocity-based adaptation (blue) and with the proposed adaptive impedance learning method (green). (a) Actual trajectories over one trial in each condition. (b) Average learned and actual disturbance. (c) Average tracking error magnitude with $\pm$ standard deviation (std) error bars. (d) Average effort of the user with $\pm$ std error bars.

disturbance, serves as a reference for best possible guidance. This reference trajectory slightly deviates from the 10 cm radius circle, due to the H-MAN sensing error from cable deformation and to the variability of exerted force. We see in Fig. 4(a) that in the control off condition the disturbance largely affects the tracking error performance so that an effective control strategy is necessary to mitigate this negative effect. Although the velocity-based impedance adaptation improves the tracking performance, there is still tracking error with this method which does not model the environment interaction. Fig. 4(c) and (d) suggests that the proposed impedance learning controller yields a better tracking performance while decreasing the effort to drive the robot for tracking the moving target. This arises as this controller accurately identifies the virtual environment. Fig. 4(b) indeed shows that the disturbance $\mathbf{d}$ is well identified.

B. Repetitive Task

In the repetitive task, one trial consists of guiding the robot during four iterations to track a target moving on a smaller circle. The target trajectory is defined as $[5 \cos(0.25\pi t) \ 5 \sin(0.25\pi t)]^T$ cm in the presence of disturbance $\mathbf{d} = [d_x \ d_y]^T$ N where $d_x = 28(\sin 160x)^2 x + 4(\sin 80x)^2 \dot{x}$ and $d_y = 28(\cos 240y)^2 y + 4(\cos 40y)^2 \dot{y}$.

In our experiment, when the human subject stopped at a position for the prescribed time, we consider that the human completed the task. Since T_i in our proposed method in (21) is the time required by the human movement in the i th iteration to complete the movement, it is computed as the time recorded by the timer of the robot achieving speed below the specified threshold minus this prescribed time. The abovementioned method of obtaining T_i is also adaptable for the experiment with Panda robot in Section VI-B.

The iterative impedance learning in (20), (21) was implemented with the parameters $\mathbf{K}_e = 0.1 \mathbf{I}_2$, $\mathbf{A} = \mathbf{I}_2$, $\mathbf{A}_v = \mathbf{A}_p = 0.05 \mathbf{I}_2$, $\eta = 1$, $\mu = 0.01$, $\bar{T} = 24$ s. The iterative learning scheme was compared with the conditions of control off and velocity-based impedance adaptation in (24) with parameters $a_1 = a_2 = 500$, $b_1 = b_2 = 2$, $c_1 = c_2 = 100$. The maneuverability and the contact stability were evaluated from the last iteration of the five trials, where the effort metric J in (25) is restricted to the last iteration's period

$$J = \int_0^{T_m} |\mathbf{f}_{h,m}^T(t) \mathbf{v}(t)| dt \quad (27)$$

where T_m is the last iteration's duration and m equals 4 for this experiment. Similarly, the learned disturbance in (26) is

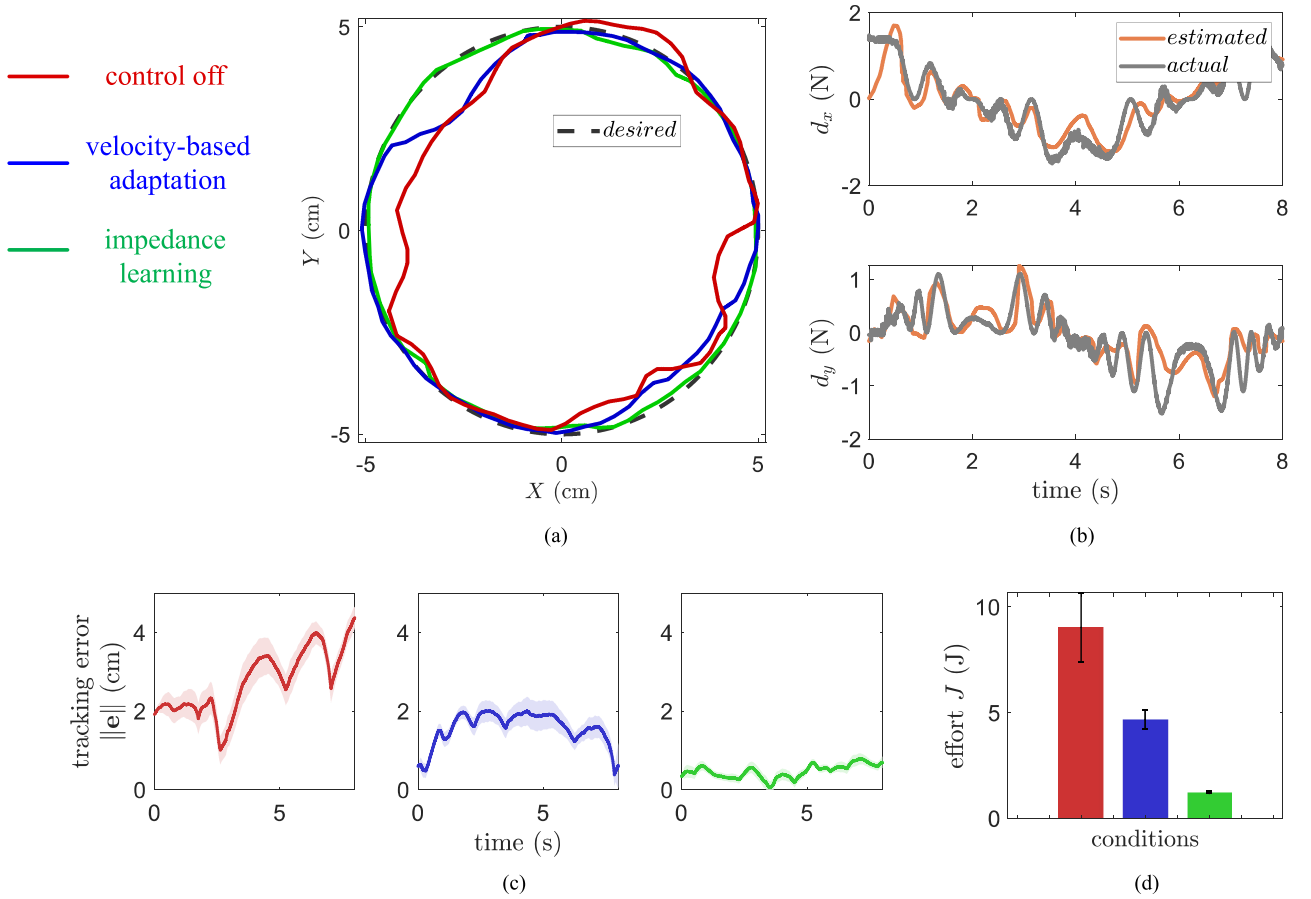


Fig. 5. Results for a repetitive task carried out by a representative subject (same subject as Fig. 4) with the H-MAN robot, in the three conditions of control off (red), with the velocity-based adaptation (blue) and with the proposed iterative impedance learning method in the last (4th) iteration (green). (a) Actual trajectories over one trial. (b) Average learned and actual disturbances. (c) Average tracking error magnitude with $\pm$ std error bars. (d) Average effort of the user with $\pm$ error std bars.

redefined as

$$\hat{d} = [\hat{d}_x \ \hat{d}_y]^T = \hat{K}_m x_m + \hat{K}_{v,m} \dot{x}_m + \hat{k}_{p,m} \quad (28)$$

where $\hat{K}_m$, $\hat{K}_{v,m}$, $\hat{k}_{p,m}$ are computed from (21).

Fig. 5(a) shows the trajectory in the three control conditions over one trial of each subject. Here also the disturbance has a significant effect on tracking performance relative to the reference trajectory (without disturbance and without control), which is partly addressed by the velocity-based adaptation. In comparison, the proposed impedance learning method reduces the tracking error [see Fig. 5(c)] to a minimum, and also minimizes the effort to drive the robot [see Fig. 5(d)]. The average learned and the actual disturbance of four iterations are illustrated in Fig. 5(b). We see that the disturbance learned by iterative impedance learning laws in (21) converges to the actual values. In comparison with the nonrepetitive tracking task above, the controller can now identify a more complex disturbance which contributes to the good performance of our proposed method.

C. Performance With Disturbance at Different Levels

An additional experiment with the H-MAN was carried out by the same subject as before in order to explore how our

proposed impedance learning method deals with different levels of disturbance in both nonrepetitive and repetitive tasks.

Performance and effort were investigated at increasing levels $L = 1, 2, 3, 4, 5$ of disturbance. For the nonrepetitive task the disturbance was (with x_1, x_2 in m and $\dot{x}_1, \dot{x}_2$ in m/s) $d = [d_x \ d_y]^T$ N where $d_x = 8(1 + 5L)x_1 + 0.4(1 + 2L)\dot{x}_1$ and $d_y = 8(1 + 5L)x_2 + 0.4(1 + 2L)\dot{x}_2$. The disturbance for the repetitive task was with $d_x = 20(1 + 3L)(\sin 10x_1)^2 \dot{x}_1 + 8(1 + L)(\sin 10x_1)^2 \dot{x}_1$ and $d_y = 20(1 + 3L)(\cos 10x_2)^2 \dot{x}_2 + 8(1 + L)(\cos 10x_2)^2 \dot{x}_2$. Five trials were carried out with each control condition, which were presented in the same pseudorandom order as before.

Tracking performance representing the contact stability was analyzed using the *total error*

$$E \equiv \int_0^T \|e(t)\| dt \quad (29)$$

where T is the nonrepetitive task duration or the duration of the last iteration for the repetitive task.

To evaluate the improvement of contact stability and maneuverability, we calculate the decrease of total error and effort with velocity-based adaptation and impedance learning relative to the

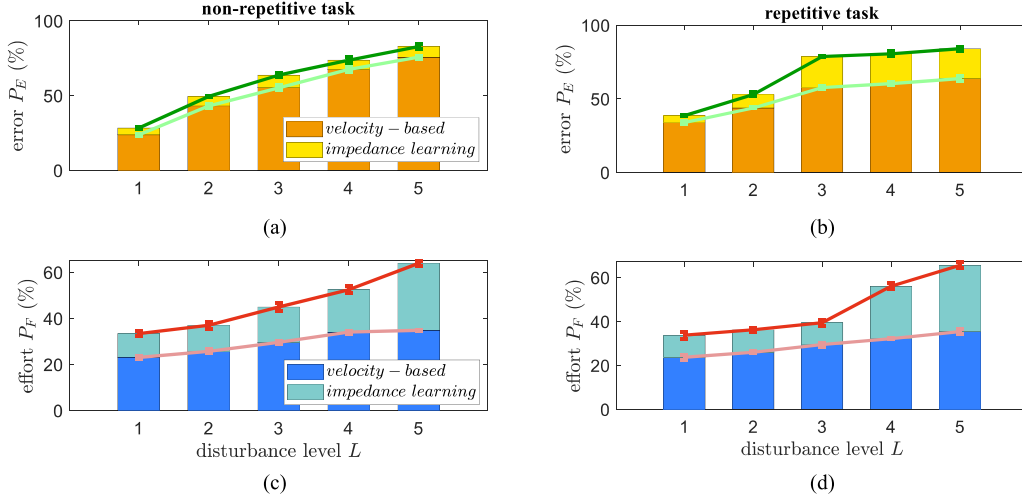


Fig. 6. Relative reduction of total tracking error and effort with disturbances of increasing magnitude (defined in the text) (a) in the nonrepetitive task and (b) in the repetitive task.

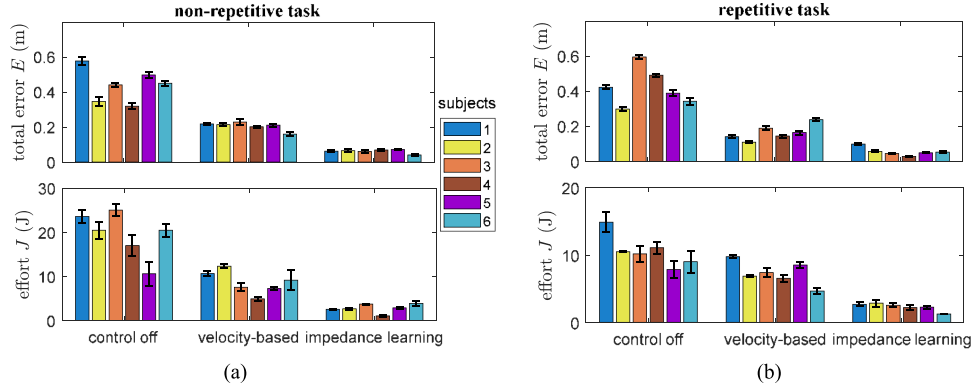


Fig. 7. Results of experiment carried out by six human subjects with the H-MAN robot and different control conditions, to validate the proposed impedance learning on a virtual disturbance. Average and $\pm$ std of total tracking error and effort are over the five trials (a) in the nonrepetitive task and (b) in the repetitive task where only the last iteration (of the five trials) is considered.

control off condition, respectively, using

$$P_E = \begin{cases} \frac{E_N - E_{ex}}{E_N} \times 100 & \text{for velocity-based adaptation} \\ \frac{E_N - E_{de}}{E_N} \times 100 & \text{for impedance learning} \end{cases} \quad (30)$$

$$P_F = \begin{cases} \frac{J_N - J_{ex}}{J_N} \times 100 & \text{for velocity-based adaptation} \\ \frac{J_N - J_{de}}{J_N} \times 100 & \text{for impedance learning} \end{cases} \quad (31)$$

where E_N , E_{ex} , and E_{de} are values of E defined in (29) in the conditions of control off, with the velocity-based adaptation and with the proposed impedance learning method, respectively; similarly, J_N , J_{ex} , and J_{de} are, respectively, the values of J [defined in (25) for nonrepetitive tasks and (27) for repetitive tasks]. According to (30), (31), P_E and P_F indicate the improvement of trajectory tracking accuracy and effort-saving using velocity-based and proposed controls, respectively. The higher value the index is, the greater improvement brought by the controller. The parameters of the velocity-based and proposed controllers were maintained as in Sections V-A-B.

Fig. 6 shows the average percentage index P_E and P_F of five trials in different tasks. As we see in Fig. 6(a) and (b),

impedance learning does not offer an obvious advantage over the velocity-based method to reduce the tracking error for an environmental disturbance of small amplitude. However, in both the nonrepetitive and repetitive tasks, the tracking performance benefits increase with the disturbance level, exhibiting less tracking error at higher levels. Due to the limited capability of controllers in improving trajectory tracking performance and the limitation of the robot platform, no further improvement over $L = 3$ is observed in Fig. 6(b).

Fig. 6 shows that the effort reduction with both the velocity-based and impedance learning control increases with the disturbance level in the nonrepetitive and repetitive tasks. However the benefits of the impedance learning approach are larger compared to the velocity-based approach.

D. Overall Results of Multiple Subjects With H-MAN Robot

We can see in Fig. 7 that the proposed control with impedance learning outperforms the velocity-based control in improving the tracking performance and minimizing effort for both nonrepetitive and repetitive tasks, which demonstrates its advantages

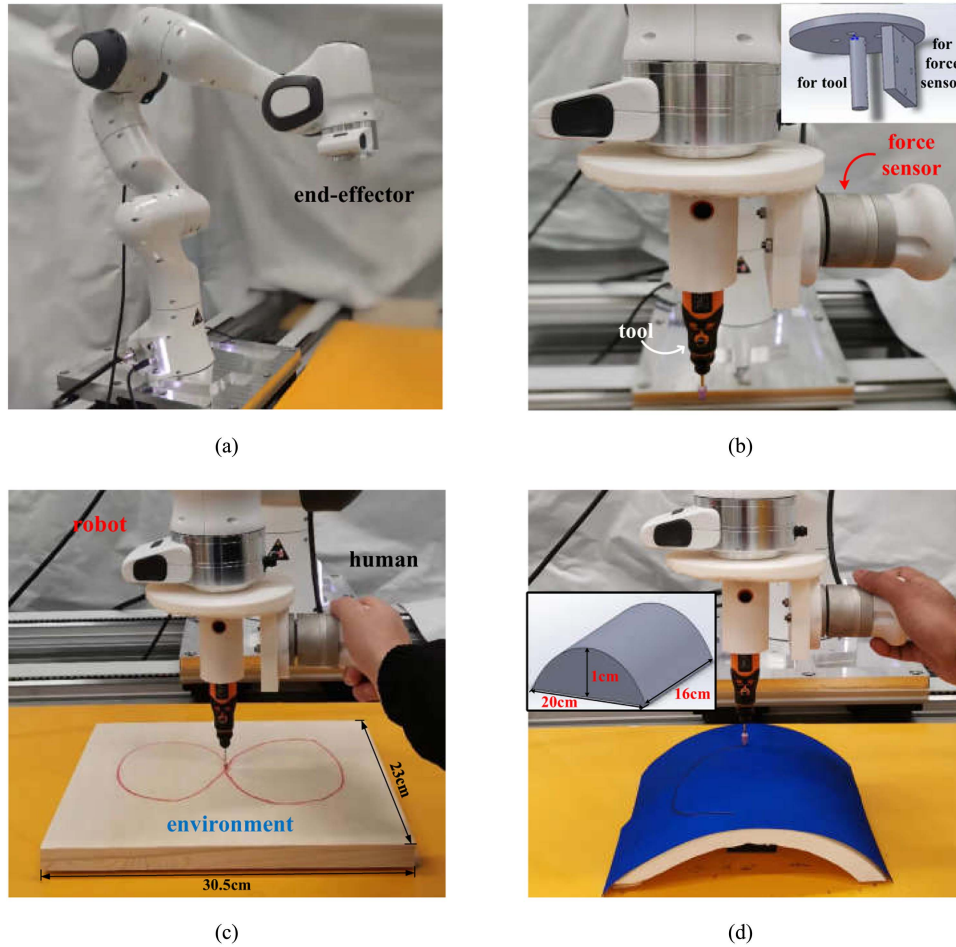


Fig. 8. Experimental setups for 3-D tasks in contact with the environment. (a) Franka Emika Panda robot with 7-DoFs used in the experiments. (b) Mechanism with handle to separate measurement of the user applied interaction force from the interaction with the environment. (c) Nonrepetitive carving task. (d) Repetitive grinding task and the CAD schematic drawing of the 3-D surface.

in ensuring the contact stability and maneuverability of the interaction system.

VI. APPLICATION TO 3-D TASKS WITH UNKNOWN ENVIRONMENT

A second experiment was carried out to test the efficiency of impedance learning control in 3-D tasks involving unknown environment's interactions, as well as its robustness to the variability of duration when humans repeat a movement. As shown in Fig. 8(a), subjects guided a 7-DoF Franka Emika Panda robot along a reference shape previously traced by the robot while interacting with a cordless rotary carving tool and a grinding tool as shown in Fig. 8(c) and (d), respectively. Compared with the software designed disturbance in Section V, the disturbance in this section arises from the contact with the environment due to the hard texture of the environment and the vibration caused by the high-frequency rotation of the tool. This experiment tests whether the interaction model of (6) is valid in this case and if the controller yields stable and efficient movement assistance to the human operator.

To decouple the human force from the environmental interaction, a mechanism was fixed between the robot extremity and the tool, as shown in Fig. 8(b). This mechanism was designed to measure the force exerted by the operator and separate it from tool interaction with the environment. The Franka control interface provides the current status of the robot and an external workstation PC was connected with the Panda robot via Ethernet to realize real-time control at 1 kHz. Joint position and velocity as well as the force exerted on the robot were recorded at 1 kHz.

The experiment protocol was approved by the UWE Research Ethics Committee (UWE REC REF No: FET-2122-59). A total of 12 human subjects without known sensorimotor impairment were recruited to carry out one of the two tasks with the Panda robot (with six subjects for each of them), who gave their informed consent prior to participation. Each subject was allowed to practice until becoming familiar with the robot guidance and the carving or grinding task. They then carried out five trials in each of the conditions control off, velocity-based control in (24) and impedance learning as described in Section V, and were naive to the conditions tested.

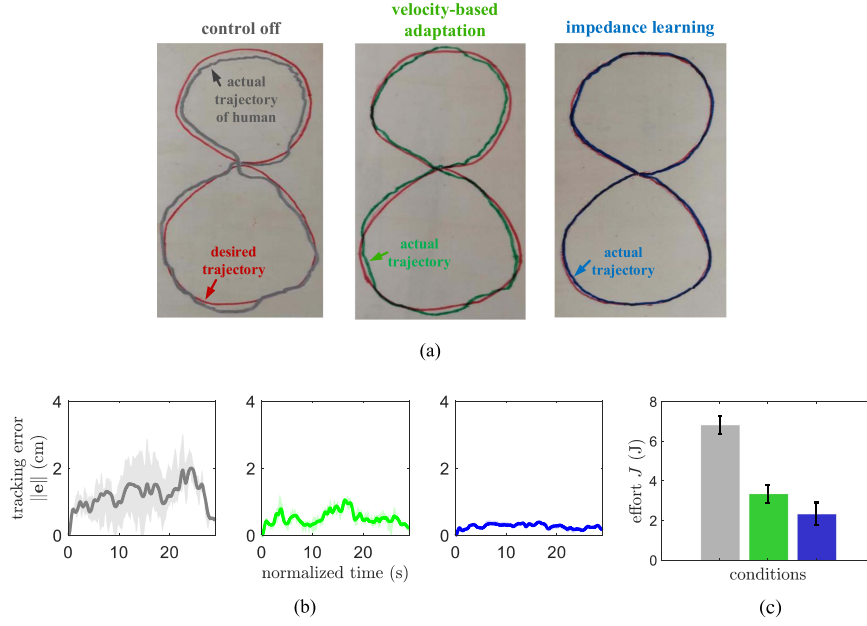


Fig. 9. Results of a representative subject in the nonrepetitive carving task with the Panda robot, in control off (grey), with velocity-based control (green), and with the proposed adaptive impedance learning method (blue). (a) “8” carved by the human user. (b) Average norm of path error with $\pm$ std error bars. (c) Average effort of the human user with $\pm$ std error bars.

Sections VI-A–B show detailed results of a representative, while the overall results of the six subjects are provided in Section VI-C.

A. Nonrepetitive Carving Task

In the nonrepetitive carving task, an “8” shape in red color was first traced by the robot on a hard wooden board with a cordless rotary carving tool, as shown in Fig. 8(c). Then, each subject guided the robot along this shape once for a trial in the three control conditions. Velocity-based adaptation was implemented with $a_1 = a_2 = 200$, $b_1 = b_2 = 10$, $c_1 = c_2 = 20$, and impedance learning with $K_e = 150I_2$, $A_f = A_d = A_c = 1000I_2$.

The time to carry out the tasks between different trials varies. To be able to compare different trials of a subject and between subjects, the errors in all trials of all subjects were normalized to the average duration of a representative subject.

Fig. 9(a) shows the character carved by a representative subject in the first trial of the three control conditions. The average norm of the path error $\|e\|$ and the average effort J of the human user defined in (25) of five trials is shown in Fig. 9(b) and (c), respectively. Both the path error and driving effort in the last trial are lowered slightly using the velocity-based control, and reduced more using impedance learning.

B. Repetitive Grinding Task

In the repetitive grinding task, a cordless rotary tool was mounted on the designed mechanism to grind a rough 3D-printed surface, as shown in Fig. 8(d). An arc was first traced by the robot on the 3-D surface as the reference for the subjects before recording. Then, each subject held the force sensor and carried out trials to grind the 3-D surface along the reference. Five trials were completed in each condition, where one trial consists of

four iterations [$m = 4$ for (27), (28)], i.e., four times back and forth. The parameters were set as $a_1 = a_2 = 200$, $b_1 = b_2 = 5$, $c_1 = c_2 = 10$ for the velocity-based adaptation and $K_e = 200I_2$, $A = A_v = 100I_2$, $A_p = 350I_2$, $\eta = 0.001$, $\mu = 0.005$, $\bar{T} = 8$ s, $a_1 = a_2 = 200$, $b_1 = b_2 = 5$, $c_1 = c_2 = 10$ for impedance learning.

The results of a representative subject are shown in Fig. 10. We see in the photos of grinding performance of Fig. 10(a) that all iterations of the first trial deviate from the reference with control off and velocity-based adaptation, without any apparent learning taking place. This bias is caused by the non-negligible robot mechanical impedance experienced when guiding the robot along the 3-D surface. To analyze how the tracking performance changes with iterations, the 3-D recorded data is reproduced in Fig. 10(b), which shows the path from the average of back and forth movements in all iterations. We see that the velocity-based method yields a slight improvement, but performance does not improve over the iterations. With iterative impedance learning, the first iteration similarly deviates from the desired path, but the movement approaches it more with each iteration.

Fig. 10(c) shows the resulting error and standard deviation error bars as a function of normalized time, in the last iteration over the five trials. The average driving effort in (27) is shown in Fig. 10(d). Overall, Fig. 10 exhibits that the velocity-based approach enables limited performance improvement, while impedance learning results in a clear decrease in both path error and the driving effort.

C. Experiment Results of Multiple Subjects for Carving and Grinding Tasks

Fig. 11(a) and (b) shows average values of total path error E defined in (29) and average effort J , respectively, over the

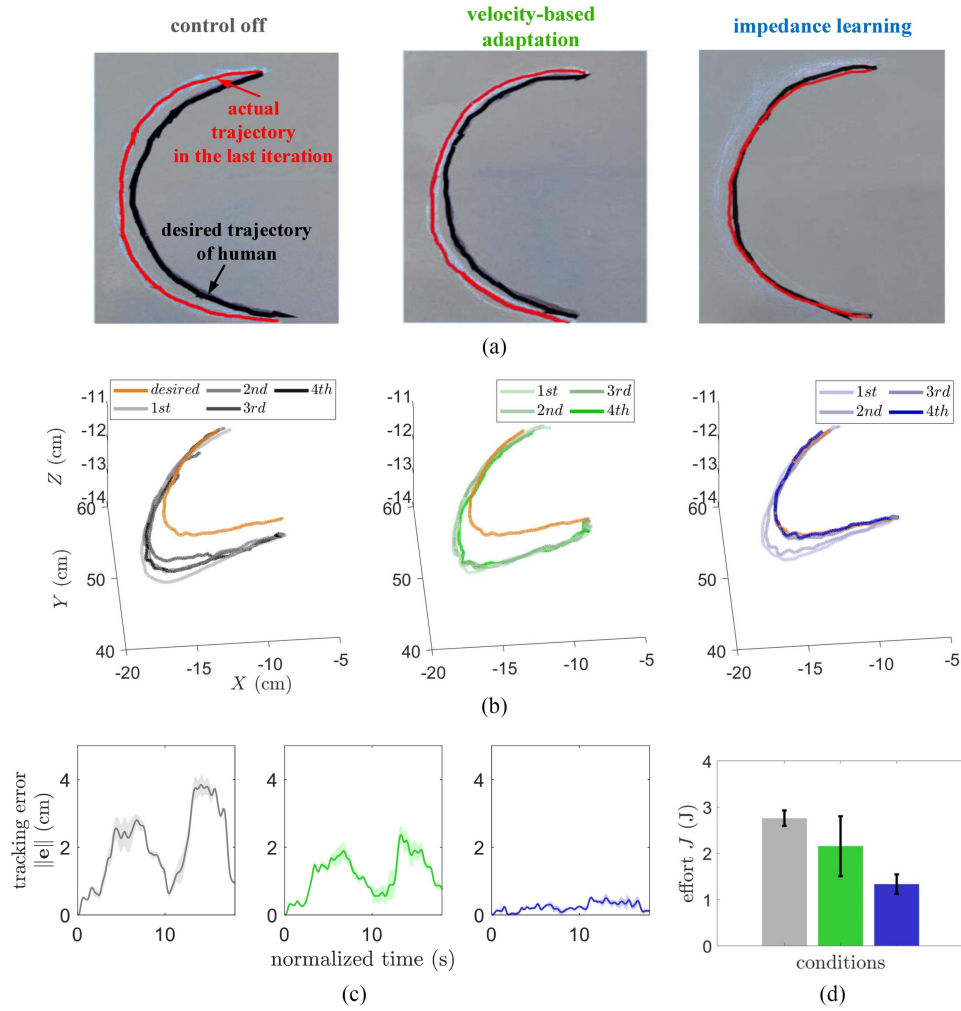


Fig. 10. Results of a representative subject (same subject as Fig. 9) in the repetitive grinding task with the Panda robot. (a) Three C-shaped trajectories grinded by the human user. (b) Actual trajectories of the human user in the 3-D space in four iterations. (c) Norm of path errors with $\pm$ std error bars. (d) Average effort of the human user with $\pm$ std error bars.

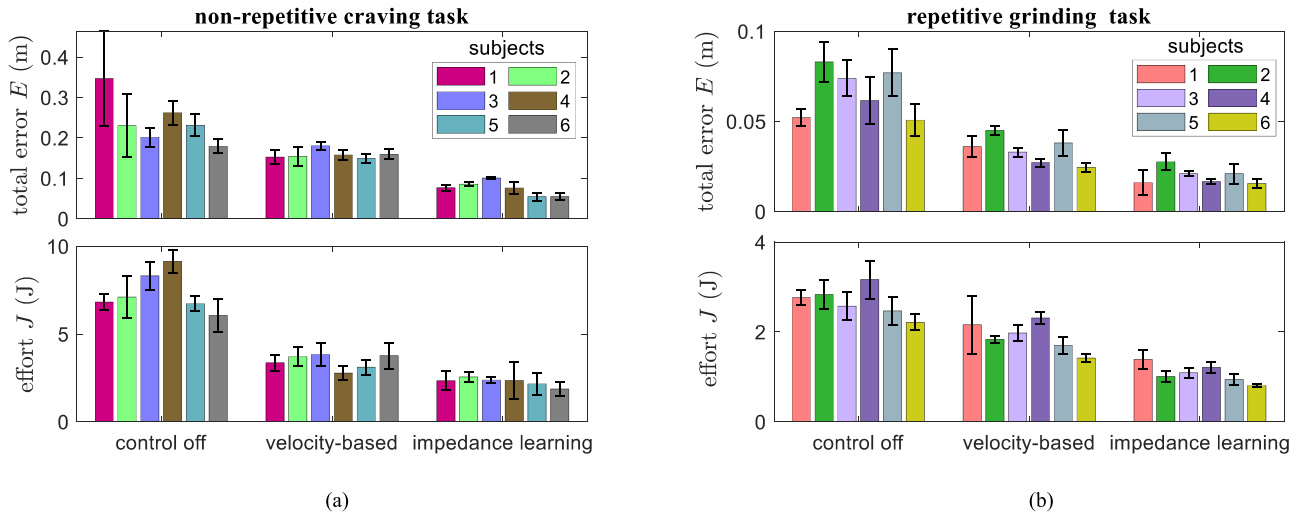


Fig. 11. Total error and effort of six subjects in the carving and grinding tasks carried out on the Panda robot with control off, velocity-based adaptation and impedance learning. Each column shows the average and $\pm$ std error bars.

six subjects that carried out each experiment. We see in Fig. 11 that the velocity-based adaptation algorithm reduces the tracking error and effort relative to control off in two tasks with a limitation as the disturbance cannot be fully compensated for. By comparison, the proposed impedance method reduces them more, exhibiting less error and effort than with velocity-based control. These results suggest that impedance learning can handle temporal variability in human movements, and ensure both contact stability and maneuverability in the presence of a real environment.

VII. DISCUSSION

For a human-guided robot, environmental disturbances can result in contact instabilities, requiring the human to increase their control effort at the expense of loss of maneuverability. In this article, to concurrently ensure contact stability and maneuverability, a novel impedance control scheme was developed. This was first done for nonrepetitive tasks, where the robot updates its impedance to adapt to the disturbance with constant coefficients. An iterative impedance control strategy was then designed to handle disturbances with varying space-related coefficients. With the proposed control, the human effort can be reduced with stable interaction and the robot can follow the desired trajectory in the presence of environmental disturbances. The larger the disturbance, the more obvious the advantage of the proposed strategy. The effectiveness of the proposed impedance control scheme for both nonrepetitive and repetitive tasks was validated in experiments to identify a known virtual interaction and to test performance in interaction with real environments.

The experiments also showed that our impedance learning controller outperforms the commonly used velocity-based impedance adaption of [4]. Similar to our method, the velocity-based adaptation method considers both interactions with the human operator and the environment. It infers the human's active intention according to human's velocity, while the robot controller cancels the effect of the environmental disturbance. However, this method does not involve any learning techniques, and the selection of its parameters requires prior knowledge about the task. Namely, the degree of the relationship between the control input and the velocity of the system should be properly reflected when setting the parameters $\mathbf{A}' = \text{diag}\{a_j\}$, $\mathbf{B}' = \text{diag}\{b_j\}$, $\mathbf{C}' = \text{diag}\{c_j\}$. For instance, parameters $\mathbf{A}'$ and $\mathbf{C}'$ should be selected according to the possible range of velocities of the task so that a variation of the damping within $[\mathbf{C}', \mathbf{A}']$ can be obtained.

In contrast, the impedance learning controller introduced in this article uses an explicit model of the environment interaction. As a consequence, it does not require prior knowledge of the environment to guarantee the stability of the system, as was proved in the appendices and verified in the experiments carried out in real environments.

The proposed controller may be extended to teleoperation, where impedance learning can ensure the contact stability between the remote robot and its environment, while no additional control effort is required from the human operator. The learning capability of the proposed controller may play a more significant

role where time delay exists in the teleoperation system. In this case, even if the human operator makes more control effort, they may still fail to handle the contact between the robot and the environment. One of our future works will be investigation of this hypothesis.

APPENDIX A PROOF OF THEOREM 1

This Appendix describes the proof of Theorem 1 in Section III. To attenuate the undesired effect of the unknown environment during trajectory tracking, both tracking error and estimation errors are minimized. We, therefore, introduce the Lyapunov function candidate

$$\begin{aligned} V &= V_1 + V_2 + V_3 + V_4 \\ V_1 &= \frac{1}{2} \dot{\mathbf{e}}^T \mathbf{M} \dot{\mathbf{e}}, \quad V_2 = \frac{1}{2} \sum_{j=1}^n \frac{1}{A_f^j} (\tilde{\mathbf{k}}^j)^T \tilde{\mathbf{k}}^j \\ V_3 &= \frac{1}{2} \sum_{j=1}^n \frac{1}{A_d^j} (\tilde{\mathbf{k}}_v^j)^T \tilde{\mathbf{k}}_v^j, \quad V_4 = \frac{1}{2} \sum_{j=1}^n \frac{1}{A_c^j} (\tilde{k}_p^j)^2 \end{aligned} \quad (32)$$

where V_1 is a quadratic function of the robot tracking error, and V_2, V_3, V_4 denote the squared estimation errors of disturbance coefficients; $\tilde{\mathbf{k}}^j, \tilde{\mathbf{k}}_v^j \in \mathbb{R}^n$ are, respectively, the j th column vectors of $\mathbb{R}^{n \times n}$ matrices $\tilde{\mathbf{K}}$ and $\tilde{\mathbf{K}}_v$, where $j = 1, 2, \dots, n$; $\tilde{k}_p^j \in \mathbb{R}$ is the j th element of $\mathbb{R}^n$ vector $\tilde{\mathbf{k}}_p$ in (18); positive constants $A_f^j, A_d^j, A_c^j \in \mathbb{R}$ are the j th diagonal elements of $\mathbb{R}^{n \times n}$ matrices $\mathbf{A}_f, \mathbf{A}_d, \mathbf{A}_c$, respectively.

Using (17) and the skew-symmetry property, i.e., $\zeta^T(\frac{1}{2}\dot{\mathbf{M}} - \mathbf{B})\zeta = 0$ for $\forall \zeta \in \mathbb{R}^n$, the derivative of V_1 is

$$\dot{V}_1 = \dot{\mathbf{e}}^T \left(-\mathbf{K}_e \dot{\mathbf{e}} + \tilde{\mathbf{K}} \mathbf{x} + \tilde{\mathbf{K}}_v \dot{\mathbf{x}} + \tilde{\mathbf{k}}_p \right). \quad (33)$$

Taking the derivatives of V_2, V_3, V_4 and using (12) yields

$$\dot{V}_2 = - \sum_{j=1}^n \frac{1}{A_f^j} (\tilde{\mathbf{k}}^j)^T \dot{\tilde{\mathbf{k}}}^j = - \sum_{j=1}^n (\tilde{\mathbf{k}}^j)^T \dot{\mathbf{e}} x_j \quad (34)$$

$$\dot{V}_3 = - \sum_{j=1}^n \frac{1}{A_d^j} (\tilde{\mathbf{k}}_v^j)^T \dot{\tilde{\mathbf{k}}}_v^j = - \sum_{j=1}^n (\tilde{\mathbf{k}}_v^j)^T \dot{\mathbf{e}} \dot{x}_j \quad (35)$$

$$\dot{V}_4 = - \sum_{j=1}^n \frac{1}{A_c^j} \tilde{k}_p^j \dot{\tilde{k}}_p^j = - \sum_{j=1}^n \tilde{k}_p^j \dot{\mathbf{e}} e_j \quad (36)$$

where $\tilde{\mathbf{k}}^j, \tilde{\mathbf{k}}_v^j \in \mathbb{R}^n$ are the j th column vectors of $\mathbb{R}^{n \times n}$ matrices $\tilde{\mathbf{K}}$ and $\tilde{\mathbf{K}}_v$, and $\tilde{k}_p^j, x_j, e_j \in \mathbb{R}$ are the j th elements of $\mathbb{R}^n$ vectors $\tilde{\mathbf{k}}_p, \mathbf{x}$, and $\mathbf{e}$.

Then, $\dot{V}$ can be computed by summing up (33)–(36) as

$$\begin{aligned} \dot{V} &= \dot{V}_1 + \dot{V}_2 + \dot{V}_3 + \dot{V}_4 \\ &= \dot{\mathbf{e}}^T \left(-\mathbf{K}_e \dot{\mathbf{e}} + \tilde{\mathbf{K}} \mathbf{x} + \tilde{\mathbf{K}}_v \dot{\mathbf{x}} + \tilde{\mathbf{k}}_p \right) - \sum_{j=1}^n \tilde{k}_p^j \dot{\mathbf{e}} e_j \\ &\quad - \sum_{j=1}^n (\tilde{\mathbf{k}}^j)^T \dot{\mathbf{e}} x_j - \sum_{j=1}^n (\tilde{\mathbf{k}}_v^j)^T \dot{\mathbf{e}} \dot{x}_j \end{aligned}$$

$$= -\dot{e}^T K_e \dot{e} \leq 0 \quad (37)$$

where the term $\dot{e}^T \tilde{K} x$ is canceled by $-\sum_{j=1}^n (\tilde{k}^j)^T \dot{e} x_j$, $\dot{e}^T \tilde{K}_v \dot{x}$ by $-\sum_{j=1}^n (\tilde{k}_v^j)^T \dot{e} \dot{x}_j$, and $\dot{e}^T \tilde{k}_p$ by $-\sum_{j=1}^n \tilde{k}_p^j \dot{e}_j$.

Applying [29, Lemma A.6], it then follows

$$\lim_{t \rightarrow \infty} \dot{e} = \lim_{t \rightarrow \infty} \ddot{e} = 0 \quad (38)$$

and with (17)

$$\lim_{t \rightarrow \infty} (\tilde{K} x + \tilde{K}_v \dot{x} + \tilde{k}_p) = 0 \quad (39)$$

so that the environmental disturbance can be estimated and the human can easily guide the robot even with the disturbance of an unknown environment. ■

APPENDIX B PROOF OF THEOREM 2

In the space domain, the robot controller in (20) can be rewritten as follows:

$$u_i = -\hat{K}_{s,i} x_i - \hat{K}_{vs,i} \dot{x}_i - \hat{k}_{ps,i} - K_e \dot{e}_i + g \quad (40)$$

where space-related impedance matrices $\hat{K}_{s,i} \in \mathbb{R}^{n \times n}$ and $\hat{K}_{vs,i} \in \mathbb{R}^{n \times n}$ and vector $\hat{k}_{ps,i} \in \mathbb{R}^n$ should be updated with the position change of the robot.

The undesired effect of the unknown environment is expected to be canceled by the robot during trajectory tracking, which is both to minimize the tracking error and improve the environment identification. Consequently, a Lyapunov function candidate is designed as follows to prove the stability of the controlled human-guided robot in Section IV

$$V_i = V_{e,i} + V_{k,i} + V_{v,i} + V_{p,i}$$

$$V_{e,i} = \frac{1}{2} \dot{e}_i^T M \dot{e}_i, \quad V_{k,i} = \frac{1}{2} \int_{\Omega} \Psi_{ks,i}(\xi) d\xi$$

$$V_{v,i} = \frac{1}{2} \int_{\Omega} \Psi_{vs,i}(\xi) d\xi, \quad V_{p,i} = \frac{1}{2} \int_{\Omega} \Psi_{ps,i}(\xi) d\xi$$

$$\varepsilon_i = \begin{cases} e_i, & \text{if } t \in [0, T_i] \\ e_i(T_i), & \text{if } t \in (T_i, T] \end{cases}$$

$$\xi = [\xi^1 \ \cdots \ \xi^n], \quad d\xi = [d\xi^1 \ \cdots \ d\xi^n]$$

$$\Psi_{ks,i}(\xi) = \left[\frac{1}{\alpha^1} \left(\tilde{k}_{s,i}^1(\xi) \right)^T \text{sgn}(v^1) \tilde{k}_{s,i}^1(\xi) \ \cdots \right. \\ \left. \frac{1}{\alpha^n} \left(\tilde{k}_{s,i}^n(\xi) \right)^T \text{sgn}(v^n) \tilde{k}_{s,i}^n(\xi) \right]$$

$$\Psi_{vs,i}(\xi) = \left[\frac{1}{\alpha_v^1} \left(\tilde{k}_{vs,i}^1(\xi) \right)^T \text{sgn}(v^1) \tilde{k}_{vs,i}^1(\xi) \ \cdots \right. \\ \left. \frac{1}{\alpha_v^n} \left(\tilde{k}_{vs,i}^n(\xi) \right)^T \text{sgn}(v^n) \tilde{k}_{vs,i}^n(\xi) \right]$$

$$\Psi_{ps,i}(\xi) = \left[\frac{1}{\alpha_p^1} \text{sgn}(v^1) \left(\tilde{k}_{ps,i}^1(\xi) \right)^2 \ \cdots \right. \\ \left. \frac{1}{\alpha_p^n} \text{sgn}(v^n) \left(\tilde{k}_{ps,i}^n(\xi) \right)^2 \right] \quad (41)$$

where $V_{e,i}$ denotes the robot tracking error; $V_{k,i}$, $V_{v,i}$, and $V_{p,i}$ are estimation errors of the coefficients of the disturbance represented by the introduced Ψ functions; $\Omega = \{x, 0 \leq t \leq T\}$; $\tilde{k}_{s,i}^j \in \mathbb{R}^n$ and $\tilde{k}_{vs,i}^j \in \mathbb{R}^n$ are the j th column vectors of $\mathbb{R}^{n \times n}$ matrices $\tilde{K}_{s,i} = K_s - \hat{K}_{s,i}$ and $\tilde{K}_{vs,i} = K_{vs} - \hat{K}_{vs,i}$, respectively; $\tilde{k}_{ps,i}^j \in \mathbb{R}$ is the j th element of $\mathbb{R}^n$ vector $\tilde{k}_{ps,i} = k_{ps} - \hat{k}_{ps,i}$; $\text{sgn}(\cdot)$ is the sign function; positive constants $\alpha^j, \alpha_v^j, \alpha_p^j \in \mathbb{R}$ are the j th diagonal elements of $\mathbb{R}^{n \times n}$ matrices A, A_v, A_p , respectively. Since the stability of the iterative learning control is verified from the perspective of iterations instead of time, $V_{k,i}$, $V_{v,i}$, and $V_{p,i}$ are, respectively, designed to be in an integral form of functions $\Psi_{ks,i}$, $\Psi_{vs,i}$, and $\Psi_{ps,i}$ in one iteration.

Since

$$V_{k,i} = \frac{1}{2} \sum_{j=1}^n \int_0^{s^j} \frac{1}{\alpha^j} \left(\tilde{k}_{s,i}^j \right)^T \text{sgn}(v^j) \tilde{k}_{s,i}^j d\xi^j \\ = \frac{1}{2} \sum_{j=1}^n \int_0^t \frac{1}{\alpha^j} \left(\tilde{k}_{s,i}^j \right)^T \text{sgn}(v^j) \tilde{k}_{s,i}^j \dot{\xi}^j d\sigma \\ = \frac{1}{2} \sum_{j=1}^n \int_0^t \frac{1}{\alpha^j} \left(\tilde{k}_{s,i}^j \right)^T |v^j| \tilde{k}_{s,i}^j d\sigma > 0 \quad (42)$$

and similarly

$$V_{v,i} = \frac{1}{2} \sum_{j=1}^n \int_0^t \frac{1}{\alpha_v^j} \left(\tilde{k}_{vs,i}^j \right)^T |v^j| \tilde{k}_{vs,i}^j d\sigma > 0 \quad (43)$$

$$V_{p,i} = \frac{1}{2} \sum_{j=1}^n \int_0^t \frac{1}{\alpha_p^j} |v^j| \left(\tilde{k}_{ps,i}^j \right)^2 d\sigma > 0 \quad (44)$$

where σ is an integration variable of time, the system Lyapunov function in (41) is positive and satisfies $V_i > 0$.

The following cases are considered for stability analysis.

Case 1: $t \in [0, T_i]$. In such a case, $\varepsilon_i = e_i$ holds and taking the derivative of $V_{e,i}$ yields

$$\dot{V}_{e,i} = \dot{e}_i^T M \ddot{e}_i + \frac{1}{2} \dot{e}_i^T \dot{M} \dot{e}_i. \quad (45)$$

Based on the skew-symmetry property [25], substituting (22) into (45) yields

$$\dot{V}_{e,i} = \dot{e}_i^T \left(-K_e \dot{e}_i + \tilde{K}_i x_i + \tilde{K}_{v,i} \dot{x}_i + \tilde{k}_{p,i} \right). \quad (46)$$

The integral of (46) can be calculated as

$$V_{e,i} = \int_0^t \dot{V}_{e,i} d\sigma + V_{e,i}(0) \\ = \int_0^t \dot{e}_i^T \left(-K_e \dot{e}_i + \tilde{K}_i x_i + \tilde{K}_{v,i} \dot{x}_i + \tilde{k}_{p,i} \right) d\sigma. \quad (47)$$

Define

$$\Delta V_{e,i} = V_{e,i} - V_{e,i-1} \quad (48)$$

and then we have

$$\Delta V_{e,i} \leq \int_0^t \dot{e}_i^T \left(-K_e \dot{e}_i + \tilde{K}_i x_i + \tilde{K}_{v,i} \dot{x}_i + \tilde{k}_{p,i} \right) d\sigma. \quad (49)$$

Similarly, we define

$$\Delta V_{k,i} = V_{k,i} - V_{k,i-1} \quad (50)$$

$$\Delta V_{v,i} = V_{v,i} - V_{v,i-1} \quad (51)$$

$$\Delta V_{p,i} = V_{p,i} - V_{p,i-1}. \quad (52)$$

Then, according to (42), we get

$$\begin{aligned} \Delta V_{k,i} = & \frac{1}{2} \sum_{j=1}^n \frac{1}{\alpha^j} \int_0^{s^j} \left[\left(\tilde{k}_{s,i}^j \right)^T \text{sgn}(v^j) \tilde{k}_{s,i}^j \right. \\ & - \left(\tilde{k}_{s,i}^j \right)^T \text{sgn}(v^j) \tilde{k}_{s,i-1}^j + \left(\tilde{k}_{s,i}^j \right)^T \text{sgn}(v^j) \tilde{k}_{s,i-1}^j \\ & \left. - \left(\tilde{k}_{s,i-1}^j \right)^T \text{sgn}(v^j) \tilde{k}_{s,i-1}^j \right] d\xi^j. \end{aligned} \quad (53)$$

Define

$$\Delta \hat{K}_{s,i}(\xi) = \hat{K}_{s,i}(\xi) - \hat{K}_{s,i-1}(\xi) \quad (54)$$

$$\Delta \hat{K}_{vs,i}(\xi) = \hat{K}_{vs,i}(\xi) - \hat{K}_{vs,i-1}(\xi) \quad (55)$$

$$\Delta \hat{k}_{ps,i}(\xi) = \hat{k}_{ps,i}(\xi) - \hat{k}_{ps,i-1}(\xi) \quad (56)$$

and thus, we have

$$\tilde{K}_{s,i}(\xi) - \tilde{K}_{s,i-1}(\xi) = -\Delta \hat{K}_{s,i}(\xi) \quad (57)$$

$$\tilde{K}_{vs,i}(\xi) - \tilde{K}_{vs,i-1}(\xi) = -\Delta \hat{K}_{vs,i}(\xi) \quad (58)$$

$$\tilde{k}_{ps,i}(\xi) - \tilde{k}_{ps,i-1}(\xi) = -\Delta \hat{k}_{ps,i}(\xi). \quad (59)$$

Using (57), (53) can be rewritten as

$$\begin{aligned} \Delta V_{k,i} = & -\frac{1}{2} \sum_{j=1}^n \frac{1}{\alpha^j} \int_0^{s^j} \left[\left(\tilde{k}_{s,i}^j \right)^T \text{sgn}(v^j) \Delta \hat{k}_{s,i}^j \right. \\ & \left. + \left(\tilde{k}_{s,i-1}^j \right)^T \text{sgn}(v^j) \Delta \hat{k}_{s,i}^j \right] d\xi^j \\ = & -\sum_{j=1}^n \frac{1}{\alpha^j} \int_0^{s^j} \left(\tilde{k}_{s,i}^j + \frac{1}{2} \Delta \hat{k}_{s,i}^j \right)^T \\ & \times \text{sgn}(v^j) \Delta \hat{k}_{s,i}^j d\xi^j \end{aligned} \quad (60)$$

and thus, one gets

$$\Delta V_{k,i} \leq -\sum_{j=1}^n \frac{1}{\alpha^j} \int_0^t \left(\tilde{k}_i^j \right)^T |v^j| \Delta \hat{k}_i^j d\sigma. \quad (61)$$

Similarly, using (43), (58) and (44), (59) yields

$$\Delta V_{v,i} \leq -\sum_{j=1}^n \frac{1}{\alpha_v^j} \int_0^t \left(\tilde{k}_{v,i}^j \right)^T |v^j| \Delta \hat{k}_{v,i}^j d\sigma \quad (62)$$

$$\Delta V_{p,i} \leq -\sum_{j=1}^n \frac{1}{\alpha_p^j} \int_0^t \tilde{k}_{p,i}^j |v^j| \Delta \hat{k}_{p,i}^j d\sigma. \quad (63)$$

Summing up inequalities (49), (61), (62), (63) yields

$$\begin{aligned} \Delta V_i \leq & \int_0^t \dot{e}_i^T \left(-K_e \dot{e}_i + \tilde{K}_i x_i + \tilde{K}_{v,i} \dot{x}_i + \tilde{k}_{p,i} \right) d\sigma \\ & - \sum_{j=1}^n \frac{1}{\alpha^j} \int_0^t \left(\tilde{k}_i^j \right)^T |v^j| \Delta \hat{k}_i^j d\sigma \\ & - \sum_{j=1}^n \frac{1}{\alpha_v^j} \int_0^t \left(\tilde{k}_{v,i}^j \right)^T |v^j| \Delta \hat{k}_{v,i}^j d\sigma \\ & - \sum_{j=1}^n \frac{1}{\alpha_p^j} \int_0^t \tilde{k}_{p,i}^j |v^j| \Delta \hat{k}_{p,i}^j d\sigma. \end{aligned} \quad (64)$$

Substituting (21) into inequality (64), we have

$$\begin{aligned} \Delta V_i \leq & \int_0^t \dot{e}_i^T \left(-K_e \dot{e}_i + \tilde{K}_i x_i + \tilde{K}_{v,i} \dot{x}_i + \tilde{k}_{p,i} \right) d\sigma \\ & - \sum_{j=1}^n \int_0^t \tilde{k}_{p,i}^j \dot{e}_i^j d\sigma - \sum_{j=1}^n \int_0^t \left(\tilde{k}_i^j \right)^T \dot{e}_i x_i^j d\sigma \\ & - \sum_{j=1}^n \int_0^t \left(\tilde{k}_{v,i}^j \right)^T \dot{e}_i x_i^j d\sigma \\ = & - \int_0^t \dot{e}_i^T K_e \dot{e}_i d\sigma \leq 0 \end{aligned} \quad (65)$$

where $e_i^j \in \mathbb{R}$ and $x_i^j \in \mathbb{R}$ are the j th elements of $e_i \in \mathbb{R}^n$ and $x_i \in \mathbb{R}^n$, respectively.

Case 2: $t \in (T_i, \bar{T}]$. According to the definition of ε_i and (47), (48), one gets

$$\begin{aligned} \Delta V_{e,i} \leq & V_{e,i}(t) \\ = & \int_0^{T_i} \dot{e}_i^T \left(-K_e \dot{e}_i + \tilde{K}_i x_i + \tilde{K}_{v,i} \dot{x}_i + \tilde{k}_{p,i} \right) d\sigma. \end{aligned} \quad (66)$$

Observing (21), it is known that $\hat{K}_i$, $\hat{K}_{v,i}$, and $\hat{k}_{p,i}$ stop updating during $t \in (T_i, \bar{T}]$ so that $\Delta \hat{K}_i = \Delta \hat{K}_{v,i} = \Delta \hat{k}_{p,i} = 0$. Then, according to (61)–(63), the following inequality holds:

$$\begin{aligned} \Delta V_{k,i} + \Delta V_{v,i} + \Delta V_{p,i} \\ \leq & -\sum_{j=1}^n \frac{1}{\alpha^j} \int_0^{T_i} \left(\tilde{k}_i^j \right)^T |v^j| \Delta \hat{k}_i^j d\sigma \\ & - \sum_{j=1}^n \frac{1}{\alpha_v^j} \int_0^{T_i} \left(\tilde{k}_{v,i}^j \right)^T |v^j| \Delta \hat{k}_{v,i}^j d\sigma \\ & - \sum_{j=1}^n \frac{1}{\alpha_p^j} \int_0^{T_i} \tilde{k}_{p,i}^j |v^j| \Delta \hat{k}_{p,i}^j d\sigma. \end{aligned} \quad (67)$$

Combining inequalities (66), (67), we have

$$\begin{aligned} \Delta V_i = & \Delta V_{e,i} + \Delta V_{k,i} + \Delta V_{v,i} + \Delta V_{p,i} \\ \leq & \int_0^{T_i} \dot{e}_i^T \left(-K_e \dot{e}_i + \tilde{K}_i x_i + \tilde{K}_{v,i} \dot{x}_i + \tilde{k}_{p,i} \right) d\sigma \end{aligned}$$

$$\begin{aligned}
& - \sum_{j=1}^n \frac{1}{\alpha_v^j} \int_0^{T_i} (\tilde{\mathbf{k}}_i^j)^T |v^j| \Delta \hat{\mathbf{k}}_i^j d\sigma \\
& - \sum_{j=1}^n \frac{1}{\alpha_v^j} \int_0^{T_i} (\tilde{\mathbf{k}}_{v,i}^j)^T |v^j| \Delta \hat{\mathbf{k}}_{v,i}^j d\sigma \\
& - \sum_{j=1}^n \frac{1}{\alpha_p^j} \int_0^{T_i} \tilde{k}_{p,i}^j |v^j| \Delta \hat{k}_{p,i}^j d\sigma. \quad (68)
\end{aligned}$$

As a result, similar to inequalities (64), (65) in case 1 with $t = T_i$, we obtain

$$\Delta V_i \leq - \int_0^{T_i} \dot{\mathbf{e}}_i^T \mathbf{K}_e \dot{\mathbf{e}}_i d\sigma \leq 0. \quad (69)$$

Combining inequalities (65), (69), we see that $\Delta V_i \leq 0$ holds in both cases 1 and 2. Consequently, as the initial energy of the human-robot system is bounded, V_i is monotonically decreasing with iterations when $\Delta V_i < 0$. Since V_i is defined to be positive, according to the definition of V_i in (41), it indicates that

$$\lim_{i \rightarrow \infty} \tilde{\mathbf{K}}_{s,i} = \lim_{i \rightarrow \infty} \tilde{\mathbf{K}}_{v,i} = \lim_{i \rightarrow \infty} \tilde{\mathbf{k}}_{p,i} = \mathbf{0}. \quad (70)$$

When $\Delta V_i = 0$, it gives rise to

$$\dot{\mathbf{e}}_i = \ddot{\mathbf{e}}_i = \mathbf{0} \quad (71)$$

thus

$$\tilde{\mathbf{K}}_i \mathbf{x}_i + \tilde{\mathbf{K}}_{v,i} \dot{\mathbf{x}}_i + \tilde{\mathbf{k}}_{p,i} = \mathbf{0} \quad (72)$$

according to (22). As a result, the disturbance can be estimated by the proposed impedance learning method and the robot can follow the desired trajectory of the human in the presence of disturbances. Theorem 2 is thus verified.

ACKNOWLEDGMENT

The authors would like to thank A. Sena and H. Godaba for their careful reading and comments on the manuscript.

REFERENCES

- [1] A. Ajoudani, N. Tsagarakis, and A. Bicchi, "Tele-impedance: Teleoperation with impedance regulation using a body-machine interface," *Int. J. Robot. Res.*, vol. 31, no. 13, pp. 1642–1656, 2012.
- [2] L. Peternel, N. Tsagarakis, and A. Ajoudani, "A human-robot co-manipulation approach based on human sensorimotor information," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 25, no. 7, pp. 811–822, Jul. 2017.
- [3] R. Ikeura and H. Inooka, "Variable impedance control of a robot for cooperation with a human," in *Proc. IEEE Int. Conf. Robot. Autom.*, 1995, vol. 3, pp. 3097–3102.
- [4] R. Ikeura, T. Moriguchi, and K. Mizutani, "Optimal variable impedance control for a robot and its application to lifting an object with a human," in *Proc. IEEE Int. Workshop Robot Hum. Interactive Commun.*, 2002, pp. 500–505.
- [5] Y. Li and S. S. Ge, "Human-robot collaboration based on motion intention estimation," *IEEE/ASME Trans. Mechatronics*, vol. 19, no. 3, pp. 1007–1014, Jun. 2014.
- [6] T. Tsumugiwa, R. Yokogawa, and K. Hara, "Variable impedance control based on estimation of human arm stiffness for human-robot cooperative calligraphic task," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2002, vol. 1, pp. 644–650.
- [7] C. Mitsantisuk, K. Ohishi, and S. Katsura, "Variable mechanical stiffness control based on human stiffness estimation," in *Proc. IEEE Int. Conf. Mechatronics*, 2011, pp. 731–736.
- [8] V. Duchaine and C. M. Gosselin, "General model of human-robot co-operation using a novel velocity based variable impedance control," in *Proc. Joint EuroHaptics Conf. Symp. Haptic Interfaces Virtual Environ. Teleoperator Syst.*, 2007, pp. 446–451.
- [9] J. R. Medina, H. Börner, S. Endo, and S. Hirche, "Impedance-based Gaussian processes for modeling human motor behavior in physical and non-physical interaction," *IEEE Trans. Biomed. Eng.*, vol. 66, no. 9, pp. 2499–2511, Sep. 2019.
- [10] K. Li, H. Zhao, T. Nuchkrua, Y. Yuan, and H. Ding, "Sparse bayesian learning-based adaptive impedance control in physical human-robot interaction," in *Proc. IEEE Int. Conf. Robot. Biomimetics*, 2018, pp. 2379–2385.
- [11] X. Xing, K. Maqsood, D. Huang, C. Yang, and Y. Li, "Iterative learning-based robotic controller with prescribed human-robot interaction force," *IEEE Trans. Autom. Sci. Eng.*, vol. 19, no. 4, pp. 3395–3408, Oct. 2022.
- [12] Y. Li et al., "A review on interaction control for contact robots through intent detection," *Prog. Biomed. Eng.*, vol. 4, no. 3, pp. 1–21, 2022.
- [13] W. He, S. S. Ge, Y. Li, E. Chew, and Y. S. Ng, "Neural network control of a rehabilitation robot by state and output feedback," *J. Intell. Robot. Syst.*, vol. 80, no. 1, pp. 15–31, 2015.
- [14] W. He, C. Xue, X. Yu, Z. Li, and C. Yang, "Admittance-based controller design for physical human-robot interaction in the constrained task space," *IEEE Trans. Autom. Sci. Eng.*, vol. 17, no. 4, pp. 1937–1949, Oct. 2020.
- [15] Y. Li, X. Zhou, J. Zhong, and X. Li, "Robotic impedance learning for robot-assisted physical training," *Front. Robot. AI*, vol. 6, 2019, Art. no. 78.
- [16] Y. Li, G. Ganesh, N. Jarrassé, S. Haddadin, A. Albu-Schaeffer, and E. Burdet, "Force, impedance, and trajectory learning for contact tooling and haptic identification," *IEEE Trans. Robot.*, vol. 34, no. 5, pp. 1170–1182, Oct. 2018.
- [17] A. Fayazi, N. Pariz, A. Karimpour, and S. H. Hosseinnia, "Robust position-based impedance control of lightweight single-link flexible robots interacting with the unknown environment via a fractional-order sliding mode controller," *Robotica*, vol. 36, pp. 1920–1942, 2018.
- [18] F. Stulp, J. Büchli, A. Ellmer, M. Mistry, E. A. Theodorou, and S. Schaal, "Model-free reinforcement learning of impedance control in stochastic environments," *IEEE Trans. Auton. Ment. Develop.*, vol. 4, no. 4, pp. 330–341, Dec. 2012.
- [19] J. Luo et al., "Reinforcement learning on variable impedance controller for high-precision robotic assembly," in *Proc. Int. Conf. Robot. Autom.*, 2019, pp. 3080–3087.
- [20] R. Martan-Martani, M. A. Lee, R. Gardner, S. Savarese, J. Bohg, and A. Garg, "Variable impedance control in end-effector space: An action space for reinforcement learning in contact-rich tasks," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2019, pp. 1010–1017.
- [21] M. Bogdanovic, M. Khadiv, and L. Righetti, "Learning variable impedance control for contact sensitive tasks," *IEEE Robot. Autom. Lett.*, vol. 5, no. 4, pp. 6129–6136, Oct. 2020.
- [22] H. Huang and C. Yang, and C. L. P. Chen, "Optimal robot-environment interaction under broad fuzzy neural adaptive control," *IEEE Trans. Cybern.*, vol. 51, no. 7, pp. 3824–3835, Jul. 2021.
- [23] E. Burdet, D. W. Franklin, and T. E. Milner, *Human Robotics: Neuromechanics and Motor Control*. Cambridge, MA, USA: MIT Press, 2013.
- [24] M. Bin Suleiman, Z. B. Binti Ibrahim, and A. F. N. Bin Rasedee, "Solution of higher-order odes using backward difference method," *Math. Problems Eng.*, vol. 2011, pp. 1–18, 2011.
- [25] J.-J. E. Slotine and W. Li, *Applied Nonlinear Control*. Englewood Cliffs, NJ, USA: Prentice-Hall, 1991.
- [26] C.-L. Chen and Y.-H. Yang, "Position-dependent disturbance rejection using spatial-based adaptive feedback linearization repetitive control," *Int. J. Robust Nonlinear Control*, vol. 19, no. 12, pp. 1337–1363, 2009.
- [27] E. Burdet, R. Osu, D. W. Franklin, T. Yoshioka, T. E. Milner, and M. Kawato, "A method for measuring endpoint stiffness during multi-joint arm movements," *J. Biomech.*, vol. 33, no. 12, pp. 1705–1709, 2000.
- [28] D. Shen and J. Xu, "Adaptive learning control for nonlinear systems with randomly varying iteration lengths," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 4, pp. 1119–1132, Apr. 2019.
- [29] M. S. D. Queiroz, D. M. Dawson, S. P. Nagarkatti, and F. Zhang, *Lyapunov-Based Control of Mechanical Systems*. Berlin, Germany: Springer, 2012.



Xueyan Xing received the B.S. degree in automation from Northeastern University, Shenyang, China, in 2015, the M.S. degree in control engineering from the Harbin Institute of Technology, Harbin, China, in 2017, and the Ph.D. degree in control theory and control engineering from Beihang University, Beijing, China, in 2020.

She was a Research Fellow with the Department of Engineering and Design, University of Sussex, Brighton, U.K., from 2020 to 2022. She is currently a Research Fellow with the Continental-NTU Corporate Lab, Nanyang Technological University, Singapore. Her current research interests include physical human–robot interaction, distributed control and optimization, and vibration control.



Etienne Burdet (Member, IEEE) received the M.S. degree in mathematics, the M.S. degree in physics, and the Ph.D. degree in robotics from ETH-Zurich, Zurich, Switzerland, in 1990, 1991, and 1996, respectively.

He is currently a Professor and a Chair of Human Robotics with Imperial College London. He uses an integrative approach of neuroscience and robotics to investigate human sensorimotor control, and to design efficient assistive devices and training systems for neuro-rehabilitation, which are tested in clinical trials and commercialized. His main research interest is in human–machine interaction.



Weiyong Si received the M.S. degree in aerospace engineering from the Beijing Institute of Technology, Beijing, China, in 2018. He is currently working toward the Ph.D. degree in robotics with Bristol Robotics Laboratory, University of the West of England, Bristol, U.K.

His research interests include robot skill learning, teleoperation, and robot control.



Chenguang Yang (Senior Member, IEEE) received the Ph.D. degree in control engineering from the National University of Singapore, Singapore, in 2010.

He received a postdoctoral training in human robotics from the Imperial College London, London, U.K. His research interest lies in human–robot interaction and intelligent system design.

Dr. Yang was the recipient of U.K. EPSRC UKRI Innovation Fellowship and individual EU Marie Curie International Incoming Fellowship. As the lead author, he was also the recipient of the IEEE Transactions on Robotics Best Paper Award (2012) and IEEE Transactions on Neural Networks and Learning Systems Outstanding Paper Award (2022). He is the Corresponding Co-Chair of IEEE Technical Committee on Collaborative Automation for Flexible Manufacturing, a Fellow of Institute of Engineering and Technology (IET), a Fellow of Institution of Mechanical Engineers (IMechE), and a Fellow of British Computer Society (BCS).



Yanan Li (Senior Member, IEEE) received the B.Eng. and M.Eng. degrees in automatic control from the Harbin Institute of Technology, Harbin, China, in 2006 and 2008, respectively, and the Ph.D. degree in robotics from the National University of Singapore, Singapore, in 2013.

From 2015 to 2017, he was a Research Associate with the Department of Bioengineering, Imperial College London, London, U.K. From 2013 to 2015, he was a Research Scientist with the Institute for Infocomm Research, Agency for Science, Technology and Research, Singapore. He is currently a Senior Lecturer in Control Engineering with the Department of Engineering and Design, University of Sussex, Brighton, U.K. His general research interests include human–robot interaction, robot control, and control theory and applications.